

Numerical Algorithms: Problem Set 2 — Complete Solutions

With Detailed Explanations for Exam Preparation

Contents

1	Part I: True/False (Questions 1–20)	2
2	Part II: Short Answer (Questions 21–50)	6
3	Part III: Algorithms and Computation (Questions 51–70)	16
4	Part IV: Long Answer (Questions 71–100)	25

1 Part I: True/False (Questions 1–20)

Key idea: For each T/F, mark and give a one-sentence justification.

Question 1

The centered difference approximation

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

has truncation error of order $O(h^2)$ when f is sufficiently smooth.

Solution: **TRUE.** Expand via Taylor series:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \dots$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \dots$$

Subtracting: $f(x+h) - f(x-h) = 2hf'(x) + \frac{h^3}{3}f'''(x) + O(h^5)$. Dividing by $2h$:

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{h^2}{6}f'''(x) + O(h^4)$$

The leading error term is $\frac{h^2}{6}f'''(x)$, so the truncation error is $O(h^2)$.

Question 2

The forward difference approximation

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

has truncation error of order $O(h^2)$ when f is sufficiently smooth.

Solution: **FALSE.** The forward difference has truncation error $O(h)$, not $O(h^2)$. From Taylor:
 $f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \dots$, so

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2}f''(x) + O(h^2).$$

The leading error is $\frac{h}{2}f''(x) = O(h)$.

Question 3

The (composite) trapezoidal rule is exact for all polynomials of degree ≤ 1 .

Solution: **TRUE.** The trapezoidal rule approximates $\int_a^b f(x) dx$ by the area of the trapezoid using linear interpolation between $f(a)$ and $f(b)$. For any linear polynomial $f(x) = c_0 + c_1x$, the linear interpolant is f itself, so the approximation is exact with zero error.

Question 4

Simpson's rule (with equally spaced points) is exact for all polynomials of degree ≤ 3 .

Solution: TRUE. Simpson's rule uses quadratic interpolation through 3 equally spaced points, giving degree of exactness 3. Although the interpolating polynomial has degree 2, the extra symmetry causes the cubic error term to cancel exactly (the integral of x^3 over a symmetric interval is 0). Hence it integrates all degree ≤ 3 polynomials exactly.

Question 5

If the objective f_0 and all inequality constraint functions f_i are convex, and all equality constraints are affine, then the problem is a convex optimization problem in standard form.

Solution: TRUE. This is precisely the definition of a convex optimization problem: convex objective, convex inequality constraints $f_i(x) \leq 0$, and affine equality constraints $h_j(x) = 0$. The feasible set (intersection of convex sets defined by convex inequalities and affine equalities) is convex.

Question 6

The Lagrange dual function $g(\lambda, \nu)$ is always convex in (λ, ν) .

Solution: FALSE. The dual function is always *concave* in (λ, ν) , not convex. It is defined as $g(\lambda, \nu) = \inf_x L(x, \lambda, \nu)$, a pointwise infimum of affine (hence concave) functions of (λ, ν) . The infimum of a family of affine functions is concave.

Question 7

Weak duality for a minimization problem says the dual optimal value d^* is always an *upper* bound on the primal optimal value p^* .

Solution: FALSE. For a minimization problem, weak duality says $d^* \leq p^*$, i.e., d^* is a *lower* bound on p^* . The dual always underestimates the primal optimum ($g(\lambda, \nu) \leq p^*$ for all dual feasible (λ, ν)).

Question 8

At a feasible point x , an inequality constraint $f_i(x) \leq 0$ is *active* if and only if $f_i(x) = 0$.

Solution: TRUE. By definition, an inequality constraint is active (or binding) at x if it holds with equality: $f_i(x) = 0$. It is inactive if $f_i(x) < 0$ (strict inequality).

Question 9

For any function f (convex or nonconvex), the biconjugate f^{**} always equals f .

Solution: FALSE. The biconjugate f^{**} equals f if and only if f is closed (lower semicontinuous) and convex. For nonconvex functions, f^{**} is the convex hull (greatest convex lower bound) of f , which is generally different from f .

Question 10

Slater's condition is sufficient (but not necessary) for strong duality in a convex optimization problem.

Solution: TRUE. Slater's condition states that there exists a strictly feasible point: $f_i(x) < 0$ for all i and $h_j(x) = 0$ for all j . If this holds for a convex problem, then strong duality holds ($d^* = p^*$). It is not necessary — strong duality can hold even when no strictly feasible point exists (e.g., in LP).

Question 11

If f is convex and differentiable on $\mathbb{R}^n$, then any point x^* satisfying $\nabla f(x^*) = 0$ is a global minimizer.

Solution: TRUE. For a convex differentiable function, the first-order condition $\nabla f(x^*) = 0$ is both necessary and sufficient for x^* to be a global minimizer. This follows from the first-order convexity condition: $f(y) \geq f(x^*) + \nabla f(x^*)^T(y - x^*) = f(x^*)$ for all y .

Question 12

Complementary slackness for a minimization problem includes the condition $\lambda_i^* f_i(x^*) = 0$ with $\lambda_i^* \geq 0$ and $f_i(x^*) \leq 0$.

Solution: TRUE. This is the complementary slackness condition: either the multiplier $\lambda_i^* = 0$ (constraint is inactive/not binding) or $f_i(x^*) = 0$ (constraint is active/binding). Both the sign condition on λ and the feasibility $f_i(x^*) \leq 0$ are part of the KKT conditions.

Question 13

Newton's method for unconstrained minimization updates as

$$x_{k+1} = x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k),$$

whenever $\nabla^2 f(x_k)$ is invertible.

Solution: TRUE. Newton's direction minimizes the second-order Taylor approximation of f around x_k :

$$f(x_k + d) \approx f(x_k) + \nabla f(x_k)^T d + \frac{1}{2} d^T \nabla^2 f(x_k) d.$$

Setting the gradient w.r.t. d to zero: $\nabla^2 f(x_k) d = -\nabla f(x_k)$, giving $d = -(\nabla^2 f(x_k))^{-1} \nabla f(x_k)$.

Question 14

Q-learning is model-free: it does not require knowledge of transition probabilities.

Solution: TRUE. Q-learning is a model-free RL algorithm that learns the action-value function $Q^*(s, a)$ directly from sampled transitions (s, a, r, s') without needing to know the transition probability function $P(s'|s, a)$ or the reward function explicitly.

Question 15

The Bellman optimality equation for Q^* can be written as

$$Q^*(s, a) = \mathbb{E}[r + \gamma \max_{a'} Q^*(s', a')].$$

Solution: **TRUE.** This is correct. The Bellman optimality equation says the optimal Q-value for (s, a) equals the expected immediate reward plus the discounted maximum Q-value achievable from the next state s' , where the expectation is over randomness in r and s' .

Question 16

Under standard conditions (sufficient exploration and appropriate learning rates), tabular Q-learning diverges.

Solution: **FALSE.** Under standard conditions (each state-action pair visited infinitely often, learning rates satisfying the Robbins–Monro conditions $\sum \alpha_t = \infty$, $\sum \alpha_t^2 < \infty$), tabular Q-learning *converges* to Q^* . Divergence can occur with function approximation (e.g., neural networks), but not in the tabular setting.

Question 17

Experience replay is used primarily to increase the size (number of parameters) of the neural network in deep Q-learning.

Solution: **FALSE.** Experience replay is used to (1) break temporal correlations between consecutive training samples and (2) improve sample efficiency by reusing past transitions. It has nothing to do with the size of the network. It stores transitions in a replay buffer and samples random minibatches for training.

Question 18

In an MDP, the relationship $V^\pi(s) = \sum_a \pi(a | s) Q^\pi(s, a)$ holds.

Solution: **TRUE.** By definition, $V^\pi(s) = \mathbb{E}[G_t | S_t = s]$ under policy π . Conditioning on the first action:

$$V^\pi(s) = \sum_a \pi(a | s) \mathbb{E}[G_t | S_t = s, A_t = a] = \sum_a \pi(a | s) Q^\pi(s, a).$$

Question 19

For smooth integrands, the global error of the composite trapezoidal rule is $O(h)$, where h is the subinterval width.

Solution: **FALSE.** The global error of the composite trapezoidal rule is $O(h^2)$, not $O(h)$. The local error on each subinterval is $O(h^3)$, and summing $n = (b - a)/h$ subintervals gives global error $O(h^3) \times O(1/h) = O(h^2)$.

Question 20

For any primal feasible x and dual feasible (λ, ν) , the duality gap $f_0(x) - g(\lambda, \nu)$ is always nonnegative.

Solution: **TRUE.** By weak duality: $g(\lambda, \nu) \leq p^* \leq f_0(x)$ for any primal feasible x and dual feasible (λ, ν) . Therefore $f_0(x) - g(\lambda, \nu) \geq 0$ always.

2 Part II: Short Answer (Questions 21–50)

Question 21

Write the **standard form** of an optimization problem. Define: optimization variable, objective function, inequality constraints, equality constraints, feasible set, and optimal value p^* .

Solution: The standard form is:

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m \\ & && h_j(x) = 0, \quad j = 1, \dots, p \end{aligned}$$

- **Optimization variable:** $x \in \mathbb{R}^n$ – the unknown to be determined.
- **Objective function:** $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ – the quantity to minimize.
- **Inequality constraints:** $f_i(x) \leq 0$ for $i = 1, \dots, m$.
- **Equality constraints:** $h_j(x) = 0$ for $j = 1, \dots, p$.
- **Feasible set:** $\mathcal{F} = \{x \in \mathbb{R}^n : f_i(x) \leq 0 \forall i, h_j(x) = 0 \forall j\}$.
- **Optimal value:** $p^* = \inf_{x \in \mathcal{F}} f_0(x)$. If $\mathcal{F} = \emptyset$, then $p^* = +\infty$; if f_0 is unbounded below on $\mathcal{F}$, then $p^* = -\infty$.

Question 22

Convert the maximization problem “maximize $f_0(x)$ subject to constraints” into an equivalent minimization problem in standard form.

Solution: Maximizing $f_0(x)$ is equivalent to minimizing $-f_0(x)$. The optimal value transforms as $p_{\min}^* = -p_{\max}^*$. The constraints remain unchanged. So:

$$\text{minimize } -f_0(x) \quad \text{subject to the same constraints.}$$

Question 23

Consider the problem minimize $f_0(x) = 1/x$ with domain $x > 0$.

- (a) What is the feasible set?
- (b) What is the optimal value p^* ?
- (c) Is the optimal value attained?

Solution:

- (a) The feasible set is $\mathcal{F} = (0, \infty)$ (no explicit constraints; the domain gives $x > 0$).
- (b) As $x \rightarrow +\infty$, $1/x \rightarrow 0^+$. So $\inf_{x>0} 1/x = 0$, and $p^* = 0$.
- (c) No – the infimum $p^* = 0$ is not attained because $1/x > 0$ for all finite $x > 0$; there is no $x \in (0, \infty)$ with $1/x = 0$. The infimum is approached but never reached.

Question 24

Consider the problem minimize $f_0(x) = -\log x$ with domain $x > 0$.

- (a) What is the feasible set?
- (b) What is the optimal value p^* ?
- (c) Is the problem bounded below?

Solution:

- (a) The feasible set is $\mathcal{F} = (0, \infty)$.
- (b) As $x \rightarrow +\infty$, $-\log x \rightarrow -\infty$. So $\inf_{x>0} (-\log x) = -\infty$, meaning $p^* = -\infty$.
- (c) No – the problem is *unbounded below*. The objective decreases without bound as $x \rightarrow \infty$.

Question 25

Define: (i) feasible point, (ii) infeasible problem, (iii) unbounded below problem, (iv) optimal set.

Solution:

- (i) **Feasible point:** A point x satisfying all constraints: $f_i(x) \leq 0$ for all i and $h_j(x) = 0$ for all j .
- (ii) **Infeasible problem:** A problem with an empty feasible set ($\mathcal{F} = \emptyset$). No point satisfies all constraints simultaneously.
- (iii) **Unbounded below problem:** A problem where $p^* = -\infty$; i.e., there exist feasible points x_k with $f_0(x_k) \rightarrow -\infty$.
- (iv) **Optimal set:** $X^* = \{x \in \mathcal{F} : f_0(x) = p^*\}$ – the set of all minimizers. May be empty (if p^* is not attained), a single point, or a set.

Question 26

Define **active**, **inactive**, and **redundant** constraints. Give a simple example of each in $\mathbb{R}^2$.

Solution: At a feasible point x :

- **Active constraint:** $f_i(x) = 0$ (the constraint is binding/tight). *Example in $\mathbb{R}^2$:* At $x^* = (1, 0)$, the constraint $x_1 \leq 1$ (i.e., $x_1 - 1 \leq 0$) is active.
- **Inactive constraint:** $f_i(x) < 0$ (strict inequality; the constraint has slack). *Example:* At $x^* = (0, 0)$, the constraint $x_1 \leq 1$ is inactive.
- **Redundant constraint:** A constraint that is never active on the feasible set (removing it doesn't change the feasible region). *Example:* The constraint $x_1 \leq 100$ is redundant if we also have $x_1 \leq 1$.

Question 27

Convert box constraints $l_i \leq x_i \leq u_i$ into standard-form inequalities $f_i(x) \leq 0$.

Solution: Each box constraint $l_i \leq x_i \leq u_i$ gives two standard-form inequalities:

$$l_i - x_i \leq 0 \quad \text{and} \quad x_i - u_i \leq 0.$$

So $f_{2i-1}(x) = l_i - x_i$ and $f_{2i}(x) = x_i - u_i$. For n variables with box constraints, we get $2n$ inequality constraints in total.

Question 28

Explain the **slack variable** transformation: how to replace $f_i(x) \leq 0$ with $f_i(x) + s_i = 0$ and $s_i \geq 0$. When is the transformed problem equivalent to the original?

Solution: Introduce a slack variable $s_i \geq 0$ and replace the inequality $f_i(x) \leq 0$ by the equality $f_i(x) + s_i = 0$. Since $s_i \geq 0$, this forces $f_i(x) = -s_i \leq 0$, which is equivalent to the original inequality.

Equivalence: The transformation is always equivalent: the feasible set in the original (x) space is the projection of the feasible set in the augmented (x, s) space onto the x -coordinates. Each feasible x in the original corresponds to exactly one $s_i = -f_i(x) \geq 0$ in the augmented problem.

This transformation is useful when algorithms require equalities (e.g., LP in standard form).

Question 29

State the definition of a **convex optimization problem in standard form**. What must be true about f_0 , each f_i , and each equality constraint?

Solution: A problem in standard form is a **convex optimization problem** if:

- The objective $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **convex** function.
- Each inequality constraint function $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **convex** function ($f_i(x) \leq 0$).
- Each equality constraint is **affine**: $h_j(x) = a_j^T x - b_j = 0$.

Under these conditions, the feasible set is convex, and any local minimum is also a global minimum.

Question 30

For the problem minimize $f_0(x)$ subject to $Ax = b$, $x \geq 0$, write the Lagrangian $L(x, \lambda, \nu)$ and state the sign restriction(s) on multipliers.

Solution: Rewrite $x \geq 0$ as $-x \leq 0$ (componentwise). The Lagrangian is:

$$L(x, \lambda, \nu) = f_0(x) + \nu^T(Ax - b) - \lambda^T x$$

where:

- $\nu \in \mathbb{R}^m$ are the **equality multipliers** (no sign restriction on ν).
- $\lambda \in \mathbb{R}^n$ are the **inequality multipliers** for $-x_i \leq 0$, and must satisfy $\lambda \geq 0$ (componentwise).

Note: The sign convention here uses $-\lambda^T x$ because the constraint is $-x \leq 0$, so multipliers $\lambda_i \geq 0$ multiply the constraint $-x_i \leq 0$.

Question 31

Define the **Lagrange dual function** $g(\lambda, \nu) = \inf_x L(x, \lambda, \nu)$. In one or two sentences, explain why g is concave.

Solution: The dual function is:

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} [f_0(x) + \sum_i \lambda_i f_i(x) + \sum_j \nu_j h_j(x)].$$

Why concave: For each fixed x , $L(x, \lambda, \nu)$ is an affine (hence concave) function of (λ, ν) . The pointwise infimum of a family of affine (concave) functions is concave. Therefore $g(\lambda, \nu)$ is concave in (λ, ν) , regardless of the convexity of the original problem.

Question 32

State **weak duality** for a minimization problem, using p^* and d^* .

Solution: Weak duality: For a minimization problem in standard form, $d^* \leq p^*$.

More precisely, for *any* dual feasible (λ, ν) (with $\lambda \geq 0$) and *any* primal feasible x :

$$g(\lambda, \nu) \leq f_0(x),$$

and therefore $d^* = \sup_{\lambda \geq 0} g(\lambda, \nu) \leq \inf_{x \text{ feasible}} f_0(x) = p^*$.

Interpretation: The dual optimal value always provides a lower bound on the primal optimal value.

Question 33

Define **strong duality** and the **duality gap**. What does it mean when the gap is zero?

Solution:

- **Duality gap:** For a given primal feasible x and dual feasible (λ, ν) , the duality gap is $f_0(x) - g(\lambda, \nu) \geq 0$.

- **Strong duality:** Strong duality holds when $d^* = p^*$ (the optimal duality gap is zero).
- **When gap is zero:** If the duality gap is zero for specific (x^*, λ^*, ν^*) , then x^* is primal optimal and (λ^*, ν^*) is dual optimal simultaneously. This is the certificate of optimality – if you find such a pair, you have solved both the primal and dual problems.

Question 34

State **Slater's condition** for a convex optimization problem with constraints $f_i(x) \leq 0$ and affine equalities.

Solution: **Slater's condition:** There exists a point $\tilde{x} \in \text{dom}(f_0) \cap \text{dom}(f_i)$ such that:

$$f_i(\tilde{x}) < 0 \quad \text{for all } i = 1, \dots, m \quad \text{and} \quad A\tilde{x} = b.$$

That is, there exists a *strictly feasible* point (all inequality constraints are strictly satisfied; equality constraints are satisfied exactly).

Consequence: If Slater's condition holds for a convex problem, then strong duality holds ($d^* = p^*$) and the dual optimum is attained.

Note: For affine inequality constraints, the strict feasibility requirement can be relaxed to just feasibility.

Question 35

Define the **Fenchel conjugate** $f^*(y)$ of $f(x)$. Compute the conjugate of $f(x) = \frac{1}{2}x^2$ (scalar case).

Solution: **Fenchel conjugate:**

$$f^*(y) = \sup_{x \in \text{dom}(f)} [y^T x - f(x)].$$

It measures the maximum “gap” between the linear function $y^T x$ and $f(x)$.

Computation for $f(x) = \frac{1}{2}x^2$:

$$f^*(y) = \sup_{x \in \mathbb{R}} [yx - \frac{1}{2}x^2].$$

This is an unconstrained quadratic in x . Taking derivative: $y - x = 0 \Rightarrow x^* = y$.

$$f^*(y) = y \cdot y - \frac{1}{2}y^2 = y^2 - \frac{1}{2}y^2 = \frac{1}{2}y^2.$$

So $f^*(y) = \frac{1}{2}y^2$. The conjugate of $\frac{1}{2}x^2$ is itself!

Question 36

Explain what is meant by **reflexivity** of the conjugate (i.e., when $f^{**} = f$). State the regularity conditions typically required.

Solution: **Reflexivity** ($f^{**} = f$): Applying the conjugate operation twice returns the original function.

The biconjugate satisfies $f^{**} \leq f$ always. Equality $f^{**} = f$ holds if and only if f is a **closed convex function** (i.e., convex and lower semicontinuous – its epigraph is a closed convex set).

Regularity conditions:

- f must be **convex**.
- f must be **lower semicontinuous** (l.s.c.): $\liminf_{x \rightarrow x_0} f(x) \geq f(x_0)$ for all x_0 .
- f must be **proper**: $f(x) > -\infty$ for all x , and $f(x) < +\infty$ for at least one x .

For nonconvex f , f^{**} gives the *convex envelope* (largest convex function $\leq f$).

Question 37

Write down the **KKT conditions** for a differentiable problem in standard form: include primal feasibility, dual feasibility, complementary slackness, and stationarity.

Solution: For the problem: minimize $f_0(x)$ s.t. $f_i(x) \leq 0$ ($i = 1, \dots, m$), $h_j(x) = 0$ ($j = 1, \dots, p$), the KKT conditions are:

1. **Primal feasibility:**

$$f_i(x^*) \leq 0, \quad i = 1, \dots, m; \quad h_j(x^*) = 0, \quad j = 1, \dots, p.$$

2. **Dual feasibility:**

$$\lambda_i^* \geq 0, \quad i = 1, \dots, m.$$

3. **Complementary slackness:**

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m.$$

4. **Stationarity:**

$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0.$$

For a convex problem satisfying Slater's condition, the KKT conditions are necessary *and sufficient* for global optimality.

Question 38

For a convex differentiable f_0 and feasible set X , state the first-order optimality condition

$$\nabla f_0(x^*)^T (y - x^*) \geq 0 \quad \text{for all } y \in X,$$

and give a geometric interpretation.

Solution: Condition: x^* is optimal if and only if $\nabla f_0(x^*)^T (y - x^*) \geq 0$ for all $y \in X$.

Geometric interpretation: The gradient $\nabla f_0(x^*)$ makes a non-negative inner product with every direction $y - x^*$ pointing into the feasible set from x^* . In other words, $-\nabla f_0(x^*)$ (the steepest descent direction) does *not* point into the feasible set – any feasible move increases f_0 (or keeps it the same). Geometrically, the feasible set X lies in the half-space where $\nabla f_0(x^*)^T (y - x^*) \geq 0$, meaning x^* is on the “downhill boundary” of X .

For the unconstrained case ($X = \mathbb{R}^n$), this reduces to $\nabla f_0(x^*) = 0$.

Question 39

Define **numerical differentiation**. What is one benefit and one drawback of choosing a very small step size h in finite difference formulas?

Solution: Numerical differentiation: Approximating derivatives of a function using function evaluations at discrete points (finite difference formulas), since exact analytical derivatives may be unavailable or expensive.

Benefit of very small h : The *truncation error* (from Taylor series terms we drop) decreases – for forward difference, truncation error $\approx \frac{h}{2}|f''(x)|$ goes to 0 as $h \rightarrow 0$.

Drawback of very small h : *Rounding (cancellation) error* increases dramatically. When h is tiny, $f(x+h) \approx f(x)$ and the subtraction $f(x+h) - f(x)$ suffers catastrophic cancellation in floating-point arithmetic. The rounding error in the difference quotient scales as $\sim \varepsilon_{\text{mach}}/h$, which blows up as $h \rightarrow 0$.

The optimal h balances these two effects: $h_{\text{opt}} \sim \sqrt{\varepsilon_{\text{mach}}}$ for forward difference, giving total error $\sim \sqrt{\varepsilon_{\text{mach}}}$.

Question 40

Write a 3-point centered difference formula for $f'(x)$ and for $f''(x)$. State the truncation error order for each.

Solution: For $f'(x)$ (3-point centered):

$$\boxed{f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}}, \quad \text{truncation error } O(h^2).$$

Derivation: from Taylor, $f(x+h) - f(x-h) = 2hf'(x) + \frac{h^3}{3}f'''(x) + O(h^5)$.

For $f''(x)$ (3-point centered):

$$\boxed{f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}}, \quad \text{truncation error } O(h^2).$$

Derivation: Add the two Taylor expansions: $f(x+h) + f(x-h) = 2f(x) + h^2f''(x) + \frac{h^4}{12}f^{(4)}(x) + O(h^6)$. Rearranging: $f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} - \frac{h^2}{12}f^{(4)}(x) + O(h^4)$.

Question 41

Define **numerical integration** (quadrature). Write the composite trapezoidal rule and composite Simpson's rule for approximating $\int_a^b f(x) dx$.

Solution: Numerical integration (quadrature): Approximating a definite integral $\int_a^b f(x) dx$ using a weighted sum of function evaluations: $\int_a^b f(x) dx \approx \sum_{i=0}^n w_i f(x_i)$.

Composite Trapezoidal Rule with n equal subintervals, $h = (b-a)/n$, $x_i = a + ih$:

$$T_n = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)] = h \left[\frac{1}{2}f(x_0) + \sum_{i=1}^{n-1} f(x_i) + \frac{1}{2}f(x_n) \right].$$

Composite Simpson's Rule (requires n even), $h = (b-a)/n$:

$$S_n = \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 4f(x_{n-1}) + f(x_n)].$$

Pattern of weights: 1, 4, 2, 4, 2, $\dots$, 4, 2, 4, 1 (alternating 4 and 2 for interior points).

Question 42

Define the **degree of exactness** of a quadrature rule. What are the degrees of exactness of trapezoidal and Simpson's rules?

Solution: Degree of exactness: A quadrature rule has degree of exactness d if it integrates all polynomials of degree $\leq d$ exactly, and there exists a polynomial of degree $d + 1$ that it does *not* integrate exactly.

- **Trapezoidal rule:** degree of exactness **1**. It integrates constants and linear functions exactly (it uses linear interpolation), but not x^2 (the integral of x^2 over $[-h, h]$ is $2h^3/3$, but the trapezoid gives $h \cdot (h^2 + h^2)/2 = h^3 \neq 2h^3/3$ in general).

Actually, on $[0, 1]$: $\int_0^1 x^2 dx = 1/3$, but $\frac{1}{2}(0^2 + 1^2) = 1/2 \neq 1/3$.

- **Simpson's rule:** degree of exactness **3**. It integrates all polynomials of degree ≤ 3 exactly (the cubic term's contribution cancels by symmetry), but not degree 4.

Question 43

Define **Richardson extrapolation**. Explain how it can improve a derivative or integral approximation.

Solution: Richardson extrapolation: Given an approximation $D(h) = D_{\text{exact}} + ch^p + O(h^{p+1})$ for some unknown constant c , compute $D(h)$ and $D(h/2)$:

$$D(h/2) = D_{\text{exact}} + c(h/2)^p + O(h^{p+1}) = D_{\text{exact}} + \frac{ch^p}{2^p} + O(h^{p+1}).$$

Multiply $D(h/2)$ by 2^p and subtract $D(h)$:

$$2^p D(h/2) - D(h) = (2^p - 1)D_{\text{exact}} + O(h^{p+1}).$$

The extrapolated estimate:

$$\tilde{D} = \frac{2^p D(h/2) - D(h)}{2^p - 1} = D_{\text{exact}} + O(h^{p+1}),$$

which has order $p + 1$ accuracy instead of order p – one order better.

Application: For the centered difference ($p = 2$), Richardson extrapolation with $p = 2$ gives a $O(h^4)$ estimate: $\tilde{D} = \frac{4D(h/2) - D(h)}{3}$.

Question 44

What is **Newton's method**? State it for (i) root finding and (ii) unconstrained minimization.

Solution: Newton's method is an iterative algorithm that uses local quadratic (second-order) information to rapidly approach a solution.

- (i) **Root finding** (solve $f(x) = 0$): At each iteration, approximate f by its tangent line at x_k and find where it crosses zero:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad f'(x_k) \neq 0.$$

- (ii) **Unconstrained minimization** (minimize $f(x)$): At each iteration, minimize the second-order Taylor approximation of f at x_k , which means solving $\nabla^2 f(x_k) d = -\nabla f(x_k)$ for the step d :

$$x_{k+1} = x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k).$$

This is equivalent to applying Newton's root-finding method to the equation $\nabla f(x) = 0$.

Question 45

Give standard assumptions under which Newton's method exhibits **local quadratic convergence**.

Solution: Newton's method for root finding (or minimization) exhibits local quadratic convergence under:

1. f is at least twice continuously differentiable (for root-finding) or three times continuously differentiable (for minimization).
2. The root x^* (or critical point) is **non-degenerate**: $f'(x^*) = 0$ but $f''(x^*) \neq 0$ (equivalently, $\nabla^2 f(x^*)$ is invertible/positive definite for minimization).
3. The starting point x_0 is **sufficiently close** to x^* .

Under these conditions, the error satisfies $|x_{k+1} - x^*| \leq C|x_k - x^*|^2$, so the number of correct decimal digits roughly doubles each iteration.

Question 46

Define the RL **return** G_t with discount factor $\gamma \in [0, 1]$. What changes conceptually when γ is close to 0 versus close to 1?

Solution: The **discounted return** is:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}.$$

- **γ close to 0 (myopic):** Only the immediate next reward matters. $G_t \approx R_{t+1}$. The agent is short-sighted and ignores long-term consequences.
- **γ close to 1 (far-sighted):** Future rewards are almost as important as immediate ones. The agent must plan ahead over long horizons and values long-term cumulative reward.

$\gamma < 1$ also ensures the sum converges for bounded rewards: $|G_t| \leq R_{\max}/(1 - \gamma)$.

Question 47

Define the **state-value** function $V^\pi(s)$ and **action-value** function $Q^\pi(s, a)$. How are they related?

Solution:

- **State-value function:** $V^\pi(s) = \mathbb{E}_\pi[G_t | S_t = s]$ – the expected discounted return starting from state s and following policy π thereafter.
- **Action-value function:** $Q^\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$ – the expected discounted return starting from state s , taking action a , then following policy π .

Relationship:

$$V^\pi(s) = \sum_a \pi(a | s) Q^\pi(s, a).$$

V^π is the expectation of Q^π over actions chosen by π . Also, $Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' | s, a) V^\pi(s')$.

Question 48

Write the **Bellman expectation equation** for $V^\pi(s)$ and the **Bellman optimality equation** for $V^*(s)$ or $Q^*(s, a)$.

Solution: Bellman expectation equation for V^π :

$$V^\pi(s) = \sum_a \pi(a | s) \sum_{s'} P(s' | s, a) [r(s, a, s') + \gamma V^\pi(s')].$$

Bellman optimality equation for V^* :

$$V^*(s) = \max_a \sum_{s'} P(s' | s, a) [r(s, a, s') + \gamma V^*(s')].$$

Bellman optimality equation for Q^* :

$$Q^*(s, a) = \sum_{s'} P(s' | s, a) [r(s, a, s') + \gamma \max_{a'} Q^*(s', a')].$$

These are fixed-point equations. For V^* and Q^* , the optimal policy is $\pi^*(s) = \arg \max_a Q^*(s, a)$.

Question 49

What is **experience replay** in deep RL? Give two reasons it can help stabilize learning.

Solution: **Experience replay:** During training, the agent stores each transition (s_t, a_t, r_t, s_{t+1}) in a *replay buffer* (circular memory of fixed size). Instead of learning from consecutive transitions online, mini-batches are sampled uniformly at random from the buffer for gradient updates.

Two reasons it stabilizes learning:

1. **Breaks temporal correlations:** Consecutive experience samples (s_t, a_t, r_t, s_{t+1}) and $(s_{t+1}, a_{t+1}, r_{t+1}, s_{t+2})$ are highly correlated. Training on correlated data leads to biased gradient estimates and instability. Random sampling from the buffer decorrelates the training data.
2. **Improved sample efficiency:** Each transition can be used multiple times for training instead of just once, making better use of expensive environment interactions.

Question 50

In Atari-style RL from pixels, how is the state commonly represented from raw frames (e.g., pre-processing and stacking), and what is the role of a CNN in that pipeline?

Solution: Preprocessing:

- **Grayscale conversion:** Convert RGB frames (3 channels) to grayscale (1 channel) to reduce dimensionality.
- **Downsampling:** Resize frames to 84×84 pixels.
- **Frame stacking:** Stack the last $k = 4$ consecutive preprocessed frames as the state. This provides temporal information (velocity, direction of moving objects) that a single frame cannot capture.

Role of CNN: The stacked frames ($84 \times 84 \times 4$ tensor) are fed into a convolutional neural network (CNN). The CNN:

- Exploits the **spatial structure** of pixel data using shared convolutional filters.
- Learns to detect spatial features (edges, objects, game elements) hierarchically.
- Outputs a feature vector that is passed to fully-connected layers to produce $Q(s, a)$ for each action a .

3 Part III: Algorithms and Computation (Questions 51–70)

Question 51

Let $f(x) = e^x$. Approximate $f'(0)$ using the forward difference with $h = 0.1$ and $h = 0.05$. Then apply Richardson extrapolation to obtain an improved estimate.

Solution: The exact answer is $f'(0) = e^0 = 1$.

Forward difference formula: $D(h) = \frac{f(h) - f(0)}{h} = \frac{e^h - 1}{h}$.

With $h = 0.1$: $e^{0.1} \approx 1.10517092$

$$D(0.1) = \frac{1.10517092 - 1}{0.1} = \frac{0.10517092}{0.1} = 1.0517092.$$

With $h = 0.05$: $e^{0.05} \approx 1.05127110$

$$D(0.05) = \frac{1.05127110 - 1}{0.05} = \frac{0.05127110}{0.05} = 1.0254220.$$

Richardson extrapolation (forward difference has $p = 1$ error, so use $2^1 = 2$):

$$\tilde{D} = \frac{2 \cdot D(0.05) - D(0.1)}{2 - 1} = 2(1.0254220) - 1.0517092 = 2.0508440 - 1.0517092 = \mathbf{0.9991348}.$$

This is much closer to the exact value of 1 (error $\approx 8.6 \times 10^{-4}$) than either individual estimate (errors ≈ 0.052 and 0.025).

Question 52

Let $f(x) = \sin x$. Approximate $f'(1)$ using the centered difference with $h = 0.2$ and $h = 0.1$. Compare the two approximations numerically.

Solution: Exact: $f'(1) = \cos(1) \approx 0.5403023$.

Centered difference: $D(h) = \frac{f(1+h) - f(1-h)}{2h}$.

With $h = 0.2$: $\sin(1.2) \approx 0.9320390$, $\sin(0.8) \approx 0.7173561$

$$D(0.2) = \frac{0.9320390 - 0.7173561}{0.4} = \frac{0.2146829}{0.4} = 0.5367073. \quad \text{Error: } |0.5403023 - 0.5367073| = 0.003595.$$

With $h = 0.1$: $\sin(1.1) \approx 0.8912073$, $\sin(0.9) \approx 0.7833269$

$$D(0.1) = \frac{0.8912073 - 0.7833269}{0.2} = \frac{0.1078804}{0.2} = 0.5394020. \quad \text{Error: } |0.5403023 - 0.5394020| = 0.000900.$$

Comparison: Halving h from 0.2 to 0.1 reduces the error by a factor of $0.003595/0.000900 \approx 4$. This is consistent with $O(h^2)$ convergence for centered differences (halving h quarters the error).

Question 53

You are given $f(0) = 1$, $f(0.1) = 1.105170$, $f(0.2) = 1.221403$. Use a 3-point forward-difference formula to approximate $f'(0)$.

Solution: The 3-point forward difference formula for $f'(x)$ at $x = 0$ with step h is derived from fitting a quadratic through $(0, f(0))$, $(h, f(h))$, $(2h, f(2h))$:

$$f'(0) \approx \frac{-3f(0) + 4f(h) - f(2h)}{2h}.$$

With $h = 0.1$:

$$f'(0) \approx \frac{-3(1) + 4(1.105170) - 1.221403}{2(0.1)} = \frac{-3 + 4.420680 - 1.221403}{0.2} = \frac{0.199277}{0.2} = \mathbf{0.99639}.$$

This is close to the true value (this data corresponds to $f(x) = e^x$, so $f'(0) = 1$). The small error is due to rounding in the given data. This formula has $O(h^2)$ truncation error.

Question 54

Approximate $f''(1)$ for $f(x) = \ln x$ using the centered difference formula with step size $h = 0.1$.

Solution: Centered difference for second derivative:

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}.$$

Compute at $x = 1$, $h = 0.1$:

$$f(1.1) = \ln(1.1) \approx 0.09531018$$

$$f(1.0) = \ln(1.0) = 0$$

$$f(0.9) = \ln(0.9) \approx -0.10536052$$

$$f''(1) \approx \frac{0.09531018 - 2(0) + (-0.10536052)}{(0.1)^2} = \frac{0.09531018 - 0.10536052}{0.01} = \frac{-0.01005034}{0.01} = \mathbf{-1.005034}.$$

Exact: $f''(x) = -1/x^2$, so $f''(1) = -1$. Our approximation -1.005034 has error $\approx 0.005 = O(h^2)$, consistent with the $O(h^2)$ accuracy of the centered second-difference formula.

Question 55

Use the composite trapezoidal rule with $n = 4$ equal subintervals to approximate $\int_0^1 \frac{1}{1+x^2} dx$.

Solution: $h = 1/4 = 0.25$. Nodes: $x_0 = 0$, $x_1 = 0.25$, $x_2 = 0.5$, $x_3 = 0.75$, $x_4 = 1$.

x_i	$f(x_i) = 1/(1+x_i^2)$	Weight
0	$1/1 = 1.000000$	$1/2$
0.25	$1/1.0625 = 0.941176$	1
0.50	$1/1.25 = 0.800000$	1
0.75	$1/1.5625 = 0.640000$	1
1.00	$1/2 = 0.500000$	$1/2$

$$T_4 = \frac{h}{2} [f_0 + 2f_1 + 2f_2 + 2f_3 + f_4] = \frac{0.25}{2} [1 + 2(0.941176) + 2(0.8) + 2(0.64) + 0.5]$$

$$= 0.125 \times [1 + 1.882353 + 1.6 + 1.28 + 0.5] = 0.125 \times 6.262353 = \mathbf{0.782794}.$$

Exact: $\int_0^1 \frac{dx}{1+x^2} = \arctan(1) = \pi/4 \approx 0.785398$. Error ≈ 0.0026 .

Question 56

Use the composite Simpson's rule with $n = 4$ equal subintervals to approximate $\int_0^1 \frac{1}{1+x^2} dx$.

Solution: Same nodes as Q55. Simpson's rule pattern: 1, 4, 2, 4, 1.

$$S_4 = \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + f_4] = \frac{0.25}{3} [1 + 4(0.941176) + 2(0.8) + 4(0.64) + 0.5]$$

$$= \frac{0.25}{3} [1 + 3.764706 + 1.6 + 2.56 + 0.5] = \frac{0.25}{3} \times 9.424706 = \frac{0.25 \times 9.424706}{3} = \frac{2.356177}{3} = \mathbf{0.785392}.$$

Exact: $\pi/4 \approx 0.785398$. Error $\approx 6 \times 10^{-6}$. Simpson's rule is far more accurate than the trapezoidal rule for this smooth integrand!

Question 57

Romberg integration: Let $I = \int_0^1 e^{-x^2} dx$. Compute $R_{1,1}$ using one trapezoid ($h = 1$), compute $R_{2,1}$ using $h = 1/2$, then compute $R_{2,2}$ via Richardson extrapolation.

Solution: Values: $f(0) = e^0 = 1$, $f(0.5) = e^{-0.25} \approx 0.778801$, $f(1) = e^{-1} \approx 0.367879$.

$R_{1,1}$ (1 trapezoid, $h = 1$):

$$R_{1,1} = \frac{1}{2} [f(0) + f(1)] = \frac{1}{2} [1 + 0.367879] = \frac{1.367879}{2} = \mathbf{0.683940}.$$

$R_{2,1}$ (2 trapezoids, $h = 0.5$):

$$R_{2,1} = \frac{0.5}{2} [f(0) + 2f(0.5) + f(1)] = 0.25 [1 + 2(0.778801) + 0.367879] = 0.25 \times 2.925481 = \mathbf{0.731370}.$$

$R_{2,2}$ (Richardson extrapolation, $p = 2$ for trapezoidal):

$$R_{2,2} = R_{2,1} + \frac{R_{2,1} - R_{1,1}}{4^1 - 1} = 0.731370 + \frac{0.731370 - 0.683940}{3} = 0.731370 + \frac{0.047430}{3} = 0.731370 + 0.015810 = \mathbf{0.747180}.$$

Exact: $I \approx 0.746824$. $R_{2,2}$ has error $\approx 3.6 \times 10^{-4}$, far better than $R_{1,1}$ (error ≈ 0.063) or $R_{2,1}$ (error ≈ 0.016).

Question 58

Use 2-point Gauss–Legendre quadrature on $[-1, 1]$ (nodes $\pm 1/\sqrt{3}$, weights 1) to approximate $\int_{-1}^1 e^x dx$.

Solution: 2-point Gauss–Legendre rule:

$$\int_{-1}^1 f(x) dx \approx 1 \cdot f\left(-\frac{1}{\sqrt{3}}\right) + 1 \cdot f\left(\frac{1}{\sqrt{3}}\right).$$

With $f(x) = e^x$ and $1/\sqrt{3} \approx 0.577350$:

$$e^{-1/\sqrt{3}} \approx e^{-0.577350} \approx 0.561491$$

$$e^{+1/\sqrt{3}} \approx e^{+0.577350} \approx 1.780741$$

$$\int_{-1}^1 e^x dx \approx 0.561491 + 1.780741 = \mathbf{2.342232}.$$

Exact: $e^1 - e^{-1} = 2.718282 - 0.367879 = 2.350403$. Error ≈ 0.0082 .

Note: 2-point G-L is exact for degree ≤ 3 polynomials. For e^x , which is not a polynomial, there is a small error.

Question 59

Perform three iterations of Newton's method to solve $x^3 - 2x - 5 = 0$ starting from $x_0 = 2$. Report x_1, x_2, x_3 .

Solution: $f(x) = x^3 - 2x - 5$, $f'(x) = 3x^2 - 2$. Newton update: $x_{k+1} = x_k - f(x_k)/f'(x_k)$.

Iteration 1: $f(2) = 8 - 4 - 5 = -1$, $f'(2) = 12 - 2 = 10$.

$$x_1 = 2 - \frac{-1}{10} = 2 + 0.1 = \mathbf{2.1}.$$

Iteration 2: $f(2.1) = (2.1)^3 - 2(2.1) - 5 = 9.261 - 4.2 - 5 = 0.061$. $f'(2.1) = 3(4.41) - 2 = 13.23 - 2 = 11.23$.

$$x_2 = 2.1 - \frac{0.061}{11.23} = 2.1 - 0.005432 = \mathbf{2.094568}.$$

Iteration 3: $(2.094568)^3 \approx 9.18948$, $f(2.094568) \approx 9.18948 - 4.18914 - 5 = 0.00034$. $f'(2.094568) = 3(2.094568)^2 - 2 \approx 3(4.38722) - 2 = 13.16165 - 2 = 11.16165$.

$$x_3 = 2.094568 - \frac{0.00034}{11.162} \approx 2.094568 - 0.0000305 \approx \mathbf{2.094551}.$$

The root $x^* \approx 2.094551$ is converged to high precision. Note the rapid (quadratic) convergence!

Question 60

Perform three Newton steps to find a local minimizer of $f(x) = x^4 - 3x^2 + 2$ starting from $x_0 = 1.5$. Report x_1, x_2, x_3 .

Solution: $f'(x) = 4x^3 - 6x$, $f''(x) = 12x^2 - 6$. Newton update: $x_{k+1} = x_k - f'(x_k)/f''(x_k)$.

Critical points: $f'(x) = 2x(2x^2 - 3) = 0 \Rightarrow x = 0$ or $x = \pm\sqrt{3/2} \approx \pm 1.22474$.

Iteration 1: $f'(1.5) = 4(3.375) - 6(1.5) = 13.5 - 9 = 4.5$. $f''(1.5) = 12(2.25) - 6 = 27 - 6 = 21$.

$$x_1 = 1.5 - 4.5/21 = 1.5 - 0.21429 = \mathbf{1.28571}.$$

Iteration 2: $f'(1.28571) = 4(2.12471) - 6(1.28571) = 8.49884 - 7.71426 = 0.78458$. $f''(1.28571) = 12(1.65306) - 6 = 19.83673 - 6 = 13.83673$.

$$x_2 = 1.28571 - 0.78458/13.83673 = 1.28571 - 0.05669 = \mathbf{1.22902}.$$

Iteration 3: $f'(1.22902) \approx 4(1.85661) - 6(1.22902) = 7.42644 - 7.37412 = 0.05232$. $f''(1.22902) \approx 12(1.51047) - 6 = 18.12564 - 6 = 12.12564$.

$$x_3 = 1.22902 - 0.05232/12.12564 = 1.22902 - 0.00431 = \mathbf{1.22471}.$$

Converging to $x^* = \sqrt{3/2} \approx 1.22474$ (local minimum with $f(x^*) = -1/4$).

Question 61

Gradient descent: minimize $f(x_1, x_2) = (x_1 - 1)^2 + 2(x_2 + 2)^2$ using constant step size $\alpha = 0.25$ from $x^{(0)} = (0, 0)$. Compute $x^{(1)}, x^{(2)}, x^{(3)}$ and $f(x^{(k)})$ for $k = 0, 1, 2, 3$.

Solution: $\nabla f = [2(x_1 - 1), 4(x_2 + 2)]$. Update: $x^{(k+1)} = x^{(k)} - 0.25\nabla f(x^{(k)})$.

$k = 0$: $x^{(0)} = (0, 0)$, $f = 1 + 8 = 9$. $\nabla f = [-2, 8]$. $x^{(1)} = (0, 0) - 0.25(-2, 8) = (0.5, -2)$.

$k = 1$: $x^{(1)} = (0.5, -2)$, $f = (0.5 - 1)^2 + 2(-2 + 2)^2 = 0.25 + 0 = 0.25$. $\nabla f = [-1, 0]$. $x^{(2)} = (0.5, -2) - 0.25(-1, 0) = (0.75, -2)$.

$k = 2$: $x^{(2)} = (0.75, -2)$, $f = (0.75 - 1)^2 + 0 = 0.0625$. $\nabla f = [-0.5, 0]$. $x^{(3)} = (0.75, -2) - 0.25(-0.5, 0) = (0.875, -2)$.

$k = 3$: $x^{(3)} = (0.875, -2)$, $f = (0.875 - 1)^2 + 0 = 0.015625$.

k	$x^{(k)}$	$f(x^{(k)})$
0	(0, 0)	9
1	(0.5, -2)	0.25
2	(0.75, -2)	0.0625
3	(0.875, -2)	0.015625

The x_2 component converges in one step (to -2); the x_1 component converges geometrically to 1.

Question 62

Backtracking line search: $f(x) = (x - 4)^2$. At $x_0 = 0$ take direction $p_0 = -f'(x_0)$. Use Armijo backtracking with $t = 1$, $\beta = 0.5$, $c = 0.1$. Show the sequence of tested t values and give the accepted t .

Solution: $f'(x) = 2(x - 4)$, $f'(0) = -8$. Direction: $p_0 = -f'(0) = 8$ (steepest descent).

$f(x_0) = f(0) = 16$. Directional derivative: $\nabla f(x_0)^T p_0 = f'(0) \cdot p_0 = (-8)(8) = -64$.

Armijo condition: $f(x_0 + tp_0) \leq f(x_0) + c \cdot t \cdot \nabla f(x_0)^T p_0$, i.e.,

$$f(8t) \leq 16 + 0.1 \cdot t \cdot (-64) = 16 - 6.4t.$$

Test $t = 1$: $f(8) = (8 - 4)^2 = 16$. Check: $16 \leq 16 - 6.4 = 9.6$? **NO**.

Reduce: $t = 0.5 \times 1 = 0.5$. $f(8 \times 0.5) = f(4) = (4 - 4)^2 = 0$. Check: $0 \leq 16 - 6.4(0.5) = 12.8$? **YES**.

Accepted: $t = 0.5$. New point: $x_1 = 0 + 0.5 \times 8 = 4$, which is the exact minimum!

Question 63

Projected gradient: minimize $f(x) = (x - 3)^2$ subject to $0 \leq x \leq 2$. Starting $x_0 = 0$, $\alpha = 0.8$, perform three projected-gradient iterations: $x_{k+1} = \Pi_{[0,2]}(x_k - \alpha f'(x_k))$.

Solution: $f'(x) = 2(x - 3)$. Projection: $\Pi_{[0,2]}(y) = \max(0, \min(2, y))$.

The unconstrained minimizer is $x^* = 3$, which is outside $[0, 2]$. The constrained minimizer is $x^* = 2$.

Iteration 1: $x_0 = 0$, $f'(0) = 2(0 - 3) = -6$.

$$x_0 - \alpha f'(x_0) = 0 - 0.8(-6) = 4.8. \quad \Pi_{[0,2]}(4.8) = 2. \quad \Rightarrow x_1 = 2.$$

Iteration 2: $x_1 = 2$, $f'(2) = 2(2 - 3) = -2$.

$$x_1 - \alpha f'(x_1) = 2 - 0.8(-2) = 2 + 1.6 = 3.6. \quad \Pi_{[0,2]}(3.6) = 2. \quad \Rightarrow x_2 = 2.$$

Iteration 3: $x_3 = 2$ (same as above).

Result: Converged immediately to $x^* = 2$ after the first step. $f(2) = 1$.

Question 64

Solve using KKT: minimize $x^2 + y^2$ subject to $x + y = 1$. Compute (x^*, y^*) and the multiplier ν^* .

Solution: **Lagrangian:** $L(x, y, \nu) = x^2 + y^2 + \nu(x + y - 1)$.

Stationarity conditions:

$$\frac{\partial L}{\partial x} = 2x + \nu = 0 \Rightarrow x = -\frac{\nu}{2},$$

$$\frac{\partial L}{\partial y} = 2y + \nu = 0 \Rightarrow y = -\frac{\nu}{2}.$$

So $x = y$. Substituting into the constraint $x + y = 1$: $2x = 1 \Rightarrow x = y = 1/2$. Then $\nu = -2x = -1$.

$$\boxed{x^* = y^* = \frac{1}{2}, \quad \nu^* = -1, \quad p^* = (1/2)^2 + (1/2)^2 = 1/2.}$$

Geometric interpretation: The minimum of $x^2 + y^2$ on the line $x + y = 1$ is the point on the line closest to the origin, which is $(1/2, 1/2)$ by symmetry.

Question 65

Duality computation: minimize x^2 subject to $x \geq 1$.

- Form the Lagrangian and dual function.
- Solve the dual problem.
- Recover the primal optimum from the dual optimum.

Solution: Write constraint as $1 - x \leq 0$.

(a) Lagrangian: $L(x, \lambda) = x^2 + \lambda(1 - x)$ with $\lambda \geq 0$. **Dual function:** $g(\lambda) = \inf_{x \in \mathbb{R}} [x^2 - \lambda x + \lambda]$. Minimize over x : $\frac{d}{dx}[x^2 - \lambda x] = 2x - \lambda = 0 \Rightarrow x^* = \lambda/2$.

$$g(\lambda) = \left(\frac{\lambda}{2}\right)^2 - \lambda \cdot \frac{\lambda}{2} + \lambda = \frac{\lambda^2}{4} - \frac{\lambda^2}{2} + \lambda = -\frac{\lambda^2}{4} + \lambda.$$

(b) Dual problem: maximize $g(\lambda) = -\lambda^2/4 + \lambda$ over $\lambda \geq 0$. $\frac{dg}{d\lambda} = -\lambda/2 + 1 = 0 \Rightarrow \lambda^* = 2$. $d^* = g(2) = -4/4 + 2 = -1 + 2 = 1$.

(c) Primal recovery: From $\lambda^* = 2$: $x^* = \lambda^*/2 = 1$. Check: $x^* = 1 \geq 1$. $p^* = (x^*)^2 = 1 = d^*$ (strong duality holds; Slater's condition satisfied for $x > 1$).

Question 66

Compute the conjugate $f^*(y)$ for $f(x) = |x|$ (scalar). Then use it to form the dual function of the constrained problem: minimize $|x|$ subject to $x \geq 1$.

Solution: Conjugate of $|x|$:

$$f^*(y) = \sup_{x \in \mathbb{R}} [yx - |x|].$$

Split by sign of x :

- $x \geq 0$: $yx - x = (y - 1)x$. If $y > 1$: $\sup = +\infty$. If $y \leq 1$: $\sup = 0$ (at $x = 0$).
- $x < 0$: $yx + x = (y + 1)x$. If $y < -1$: $\sup = +\infty$ (as $x \rightarrow -\infty$). If $y \geq -1$: $\sup = 0$.

Therefore:

$$f^*(y) = \begin{cases} 0 & \text{if } |y| \leq 1 \\ +\infty & \text{if } |y| > 1. \end{cases}$$

This is the indicator function of $[-1, 1]$.

Dual of minimize $|x|$ subject to $x \geq 1$: Write as $1 - x \leq 0$. $L(x, \lambda) = |x| + \lambda(1 - x)$, $\lambda \geq 0$.

$$g(\lambda) = \inf_x [|x| - \lambda x] + \lambda = -\sup_x [\lambda x - |x|] + \lambda = -f^*(\lambda) + \lambda.$$

Since $f^*(\lambda) = 0$ for $|\lambda| \leq 1$ and $+\infty$ otherwise:

$$g(\lambda) = \begin{cases} \lambda & \text{if } 0 \leq \lambda \leq 1 \\ -\infty & \text{if } \lambda > 1. \end{cases}$$

Dual problem: maximize $g(\lambda) = \lambda$ subject to $0 \leq \lambda \leq 1$. Solution: $\lambda^* = 1$, $d^* = 1 = p^*$ (primal: $x^* = 1$, $|x^*| = 1$). Strong duality holds.

Question 67

Value iteration: MDP with states $\{0, 1\}$, actions $\{a, b\}$. State 0: action $a \rightarrow$ state 0, reward 0; action $b \rightarrow$ state 1, reward 1. State 1: both actions $\rightarrow$ state 1, reward 2. Let $\gamma = 0.9$, $V_0(0) = V_0(1) = 0$. Report V_1, V_2, V_3 .

Solution: Value iteration: $V_{k+1}(s) = \max_a [r(s, a) + \gamma V_k(s')]$.

Iteration 1:

$$V_1(0) = \max\{0 + 0.9 \cdot 0, 1 + 0.9 \cdot 0\} = \max\{0, 1\} = 1. \quad (\text{optimal: action } b)$$

$$V_1(1) = \max\{2 + 0.9 \cdot 0, 2 + 0.9 \cdot 0\} = 2.$$

Iteration 2:

$$V_2(0) = \max\{0 + 0.9 \cdot V_1(0), 1 + 0.9 \cdot V_1(1)\} = \max\{0.9, 1 + 1.8\} = \max\{0.9, 2.8\} = 2.8.$$

$$V_2(1) = 2 + 0.9 \cdot V_1(1) = 2 + 1.8 = 3.8.$$

Iteration 3:

$$V_3(0) = \max\{0 + 0.9(2.8), 1 + 0.9(3.8)\} = \max\{2.52, 1 + 3.42\} = \max\{2.52, 4.42\} = 4.42.$$

$$V_3(1) = 2 + 0.9(3.8) = 2 + 3.42 = 5.42.$$

True optimal (geometric series, always play b): $V^*(0) = 1 + 0.9 \cdot 2 + 0.9^2 \cdot 2 + \dots = 1 + 2\gamma/(1-\gamma) = 1 + 18 = 19$; $V^*(1) = 2/(1-0.9) = 20$. Value iteration will converge there.

Question 68

Q-learning updates (no experience replay). Use the MDP from Q67. Initialize $Q_0(s, a) = 0$, $\alpha = 0.5$, $\gamma = 0.9$. Observe transitions $(0, b, 1, 1)$, $(1, a, 2, 1)$, $(1, b, 2, 1)$. Report updated values.

Solution: Q-learning update: $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$.

Transition 1: $(s = 0, a = b, r = 1, s' = 1)$

$$Q(0, b) \leftarrow 0 + 0.5[1 + 0.9 \cdot \max(Q(1, a), Q(1, b)) - 0] = 0.5[1 + 0] = \mathbf{0.5}.$$

Transition 2: $(s = 1, a = a, r = 2, s' = 1)$

$$Q(1, a) \leftarrow 0 + 0.5[2 + 0.9 \cdot \max(Q(1, a), Q(1, b)) - 0] = 0.5[2 + 0] = \mathbf{1.0}.$$

Transition 3: $(s = 1, a = b, r = 2, s' = 1)$. Now $\max(Q(1, a), Q(1, b)) = \max(1.0, 0) = 1.0$.

$$Q(1, b) \leftarrow 0 + 0.5[2 + 0.9(1.0) - 0] = 0.5[2.9] = \mathbf{1.45}.$$

Final Q table:

(s, a)	a	b
State 0	0.0	0.5
State 1	1.0	1.45

Question 69

Solve the primal LP and its dual: minimize $3x_1 + 2x_2$ s.t. $x_1 + x_2 \geq 1, x_1, x_2 \geq 0$.

- Write a standard-form equivalent if needed.
- Write the dual.
- Solve both and verify complementary slackness.

Solution: (a) **Standard form** (introduce surplus variable $s \geq 0$): minimize $3x_1 + 2x_2$, s.t. $x_1 + x_2 - s = 1, x_1, x_2, s \geq 0$.

(b) **Dual.** Primal: $\min c^T x, Ax \geq b, x \geq 0$ with $c = [3, 2]^T, A = [1, 1], b = 1$. Dual: $\max b^T y, A^T y \leq c, y \geq 0$:

$$\text{maximize } y \quad \text{s.t.} \quad y \leq 3, y \leq 2, y \geq 0, \quad \text{i.e.,} \quad 0 \leq y \leq 2.$$

(c) **Solutions:**

Dual: $\max y$ over $[0, 2]$: $y^* = 2, d^* = 2$.

Primal: Minimize $3x_1 + 2x_2$ on $x_1 + x_2 \geq 1, x_1, x_2 \geq 0$. Since x_2 has smaller cost, set $x_1^* = 0, x_2^* = 1$: $p^* = 2$.

$p^* = d^* = 2$ (strong duality).

Complementary slackness:

- $y^*(x_1^* + x_2^* - 1) = 2(0 + 1 - 1) = 0$ (constraint active).
- $x_1^*(3 - y^*) = 0 \cdot (3 - 2) = 0$ (since $x_1^* = 0$).
- $x_2^*(2 - y^*) = 1 \cdot (2 - 2) = 0$ (dual constraint active for x_2).

Question 70

Water-filling style KKT: maximize $\log(\alpha_1 + x_1) + \log(\alpha_2 + x_2)$ s.t. $x_1, x_2 \geq 0, x_1 + x_2 = 1$, with $\alpha_1 = 0.2, \alpha_2 = 1.0$. Use KKT to compute (x_1^*, x_2^*) .

Solution: This is a concave maximization (equivalently minimize $-\log(\cdot + \cdot) - \log(\cdot + \cdot)$) subject to convex constraints – a convex problem.

Lagrangian: $L = \log(0.2 + x_1) + \log(1 + x_2) - \nu(x_1 + x_2 - 1) + \mu_1 x_1 + \mu_2 x_2$.

KKT stationarity:

$$\frac{1}{0.2 + x_1} - \nu + \mu_1 = 0, \quad \frac{1}{1 + x_2} - \nu + \mu_2 = 0.$$

Try interior solution ($x_1, x_2 > 0$, so $\mu_1 = \mu_2 = 0$):

$$\frac{1}{0.2 + x_1} = \nu = \frac{1}{1 + x_2} \implies 0.2 + x_1 = 1 + x_2 \implies x_1 - x_2 = 0.8.$$

Combined with $x_1 + x_2 = 1$: adding gives $2x_1 = 1.8$, so

$$x_1^* = 0.9, \quad x_2^* = 0.1, \quad \nu^* = \frac{1}{1.1} \approx 0.909.$$

Interpretation: This is the “water-filling” solution – allocate more resource to the channel with lower initial level ($\alpha_1 = 0.2 < \alpha_2 = 1.0$) to equalize marginal returns.

4 Part IV: Long Answer (Questions 71–100)

Question 71 — Finite Differences via Taylor Series

(a) Using Taylor expansions, derive the forward-difference and backward-difference approximations for $f'(x)$ and identify the leading-order truncation error term.

Solution: **Forward difference:** Expand $f(x+h)$ about x :

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \dots$$

Rearranging:

$$f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{h}{2}f''(x) - \dots = \frac{f(x+h) - f(x)}{h} + O(h).$$

Leading error: $-\frac{h}{2}f''(x)$.

Backward difference: Expand $f(x-h)$ about x :

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \dots$$

$$f'(x) = \frac{f(x) - f(x-h)}{h} - \frac{h}{2}f''(x) + \dots = \frac{f(x) - f(x-h)}{h} + O(h).$$

Leading error: $+\frac{h}{2}f''(x)$.

Both have $O(h)$ truncation error. f must be at least C^2 .

(b) Derive the centered-difference approximation for $f'(x)$ and show its truncation error is $O(h^2)$.

Solution: Subtract the backward Taylor expansion from the forward one:

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{2h^3}{6}f'''(x) + \dots = 2hf'(x) + \frac{h^3}{3}f'''(x) + O(h^5).$$

Dividing by $2h$:

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{h^2}{6}f'''(x) + O(h^4).$$

The truncation error is $\frac{h^2}{6}f'''(x) = O(h^2)$, one order better than forward/backward differences.

(c) Derive a centered-difference approximation for $f''(x)$ and state its truncation error order.

Solution: Add the forward and backward Taylor expansions:

$$f(x+h) + f(x-h) = 2f(x) + h^2f''(x) + \frac{h^4}{12}f^{(4)}(x) + O(h^6).$$

Rearranging:

$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} - \frac{h^2}{12}f^{(4)}(x) + O(h^4).$$

Truncation error: $O(h^2)$.

(d) State the smoothness assumptions on f required for each derivation.

Solution:

- **(a) Forward/backward difference:** $f \in C^2$ (twice continuously differentiable) to bound the $O(h)$ error term.
- **(b) Centered difference for f' :** $f \in C^3$ (three times continuously differentiable) to bound the $O(h^2)$ error term.
- **(c) Centered difference for f'' :** $f \in C^4$ (four times continuously differentiable) to bound the $O(h^2)$ error term.

Question 72 — Truncation vs. Rounding Error

(a) For the forward difference $D(h) = \frac{f(x+h)-f(x)}{h}$, model the total error as a sum of truncation error and rounding error.

Solution: Truncation error: From Taylor series, $D(h) = f'(x) + \frac{h}{2}f''(x) + \dots$, so the truncation error is approximately $|E_{\text{trunc}}| \approx \frac{h}{2}|f''(x)|$.

Rounding error: In floating-point arithmetic, $f(x+h)$ and $f(x)$ are each evaluated with relative error $\varepsilon_{\text{mach}}$, so absolute errors $\approx \varepsilon_{\text{mach}}|f|$. The difference $f(x+h) - f(x)$ has absolute error $\approx 2\varepsilon_{\text{mach}}|f|$. Dividing by h :

$$|E_{\text{round}}| \approx \frac{2\varepsilon_{\text{mach}}|f|}{h}.$$

Total error model:

$$|E_{\text{total}}| \approx \frac{h}{2}|f''(x)| + \frac{2\varepsilon_{\text{mach}}|f(x)|}{h}.$$

(b) Show the total error can be approximated by $C_1h + C_2\varepsilon_{\text{mach}}/h$ for suitable constants C_1, C_2 .

Solution: Setting $C_1 = \frac{1}{2}|f''(x)|$ and $C_2 = 2|f(x)|$, the total error model is exactly:

$$|E_{\text{total}}| \approx C_1h + \frac{C_2\varepsilon_{\text{mach}}}{h}.$$

C_1 depends on the curvature of f and C_2 on the magnitude of f .

(c) Find the h that minimizes this model of total error and interpret the result.

Solution: Minimize $\phi(h) = C_1h + C_2\varepsilon_{\text{mach}}h^{-1}$ over $h > 0$:

$$\frac{d\phi}{dh} = C_1 - \frac{C_2\varepsilon_{\text{mach}}}{h^2} = 0 \implies h_{\text{opt}} = \sqrt{\frac{C_2\varepsilon_{\text{mach}}}{C_1}}.$$

With $C_1 \sim C_2 \sim O(1)$: $h_{\text{opt}} \sim \sqrt{\varepsilon_{\text{mach}}} \approx 10^{-8}$ (for double precision $\varepsilon_{\text{mach}} \approx 10^{-16}$).

The minimum total error is:

$$\phi(h_{\text{opt}}) = 2\sqrt{C_1C_2\varepsilon_{\text{mach}}} \sim O(\sqrt{\varepsilon_{\text{mach}}}) \approx 10^{-8}.$$

Interpretation: There is an optimal step size that balances truncation and rounding errors. Smaller is not always better – below h_{opt} , rounding error dominates.

Question 73 — Richardson Extrapolation

(a) Suppose $D(h) = f'(x) + ch^p + O(h^{p+1})$. Derive an extrapolated estimate $\tilde{D}$ that cancels the ch^p term using $D(h)$ and $D(h/2)$.

Solution: Write:

$$\begin{aligned} D(h) &= f'(x) + ch^p + O(h^{p+1}) \\ D(h/2) &= f'(x) + c\left(\frac{h}{2}\right)^p + O(h^{p+1}) = f'(x) + \frac{ch^p}{2^p} + O(h^{p+1}). \end{aligned}$$

Multiply the second equation by 2^p :

$$2^p D(h/2) = 2^p f'(x) + ch^p + O(h^{p+1}).$$

Subtract $D(h)$:

$$2^p D(h/2) - D(h) = (2^p - 1)f'(x) + O(h^{p+1}).$$

Divide by $2^p - 1$:

$$\tilde{D} = \frac{2^p D(h/2) - D(h)}{2^p - 1} = f'(x) + O(h^{p+1}).$$

(b) Show that the extrapolated method has higher order accuracy.

Solution: The original method $D(h)$ has error $O(h^p)$. The extrapolated method $\tilde{D}$ has error $O(h^{p+1})$, which is one order higher. Specifically, the leading error term ch^p has been completely eliminated by taking the right linear combination of $D(h)$ and $D(h/2)$.

(c) Apply your result to the centered difference method for $f'(x)$ (identify p).

Solution: For centered differences: $D(h) = f'(x) + \frac{h^2}{6}f'''(x) + O(h^4)$, so $p = 2$ and $c = f'''(x)/6$. Applying Richardson extrapolation with $p = 2$:

$$\tilde{D} = \frac{4D(h/2) - D(h)}{4 - 1} = \frac{4D(h/2) - D(h)}{3} = f'(x) + O(h^4).$$

The extrapolated estimate achieves $O(h^4)$ accuracy using the same function evaluations, compared to $O(h^2)$ for the standard centered difference.

Question 74 — Composite Trapezoidal Rule

(a) Derive the trapezoidal rule on $[a, b]$ from linear interpolation of f .

Solution: Linearly interpolate f between $(a, f(a))$ and $(b, f(b))$:

$$p_1(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

Integrate:

$$\int_a^b p_1(x) dx = f(a)(b - a) + \frac{f(b) - f(a)}{b - a} \cdot \frac{(b - a)^2}{2} = \frac{b - a}{2} [f(a) + f(b)].$$

This is the (single) trapezoidal rule – the area of the trapezoid with vertices at $(a, 0)$, $(b, 0)$, $(b, f(b))$, $(a, f(a))$.

(b) Derive the composite trapezoidal rule with n equal subintervals.

Solution: Partition $[a, b]$ into n subintervals: $h = (b - a)/n$, $x_i = a + ih$ for $i = 0, \dots, n$. Apply the single trapezoidal rule on each $[x_{i-1}, x_i]$:

$$\int_{x_{i-1}}^{x_i} f(x) dx \approx \frac{h}{2} [f(x_{i-1}) + f(x_i)].$$

Sum over all subintervals:

$$T_n = \sum_{i=1}^n \frac{h}{2} [f(x_{i-1}) + f(x_i)] = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right].$$

Each interior node x_i ($1 \leq i \leq n-1$) appears in two consecutive trapezoids, contributing a factor of 2.

(c) Prove the rule is exact for all polynomials of degree ≤ 1 .

Solution: For a polynomial $f(x) = c_0 + c_1x$ of degree ≤ 1 , the linear interpolant $p_1(x)$ through any two points $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$ is f itself (since f is already linear). Therefore, on each subinterval:

$$\int_{x_{i-1}}^{x_i} f(x) dx = \int_{x_{i-1}}^{x_i} p_1(x) dx = \frac{h}{2} [f(x_{i-1}) + f(x_i)].$$

This is exact with zero error, so the composite rule is also exact. $\square$

(d) Derive an error bound and show the global error scales as $O(h^2)$.

Solution: On each subinterval $[x_{i-1}, x_i]$, the local error is:

$$e_i = \int_{x_{i-1}}^{x_i} f(x) dx - \frac{h}{2} [f(x_{i-1}) + f(x_i)] = -\frac{h^3}{12} f''(\xi_i)$$

for some $\xi_i \in (x_{i-1}, x_i)$ (by the Euler–Maclaurin formula / Taylor expansion).

Global error:

$$E_n = \sum_{i=1}^n e_i = -\frac{h^3}{12} \sum_{i=1}^n f''(\xi_i) \approx -\frac{h^3}{12} \cdot \frac{b-a}{h} \cdot f''(\xi) = -\frac{(b-a)h^2}{12} f''(\xi)$$

for some $\xi \in (a, b)$ (by the mean value theorem for sums).

Error bound:

$$|E_n| \leq \frac{(b-a)h^2}{12} \max_{x \in [a,b]} |f''(x)| = O(h^2).$$

The global error is $O(h^2)$, so the method converges as $h \rightarrow 0$.

Question 75 — Simpson's Rule

(a) Derive Simpson's rule on $[a, b]$ using quadratic interpolation through three equally spaced points.

Solution: Let $h = (b - a)/2$, $x_0 = a$, $x_1 = (a + b)/2$, $x_2 = b$. Fit a quadratic $p_2(x)$ through (x_0, f_0) , (x_1, f_1) , (x_2, f_2) . Using the change of variable $t = x - x_1$ (so $t \in [-h, h]$):

By Newton's forward difference or Lagrange interpolation, one can show:

$$\int_{-h}^h p_2(t) dt = \int_{-h}^h \left[f_1 + \frac{f_2 - f_0}{2h} t + \frac{f_2 - 2f_1 + f_0}{2h^2} t^2 \right] dt.$$

The odd term integrates to 0 by symmetry, and:

$$= 2hf_1 + \frac{f_2 - 2f_1 + f_0}{2h^2} \cdot \frac{2h^3}{3} = 2hf_1 + \frac{h}{3}(f_2 - 2f_1 + f_0) = \frac{h}{3}(f_0 + 4f_1 + f_2).$$

Simpson's rule:

$$\int_a^b f(x) dx \approx \frac{b-a}{6} [f(a) + 4f(\frac{a+b}{2}) + f(b)] = \frac{h}{3}[f_0 + 4f_1 + f_2].$$

(b) Show Simpson's rule is exact for all polynomials of degree ≤ 3 .

Solution: It suffices to check for basis polynomials $\{1, x, x^2, x^3\}$ on $[-h, h]$ (by symmetry/translation).

The quadratic interpolant through x_0, x_1, x_2 reproduces any polynomial of degree ≤ 2 exactly.

For $f(x) = x^3$ on $[-h, h]$ with $x_0 = -h, x_1 = 0, x_2 = h$:

Exact integral: $\int_{-h}^h x^3 dx = 0$ (odd function over symmetric interval).

Simpson's rule: $\frac{h}{3}[(-h)^3 + 4(0)^3 + h^3] = \frac{h}{3}[-h^3 + 0 + h^3] = 0$.

Since the rule is exact for degree 0, 1, 2 (quadratic interpolant is exact), and for degree 3 by symmetry, it is exact for all degree ≤ 3 polynomials. $\square$

(c) Derive the error expression involving $f^{(4)}(\xi)$.

Solution: On a single interval $[a, b]$ with $h = (b-a)/2$, the error in Simpson's rule is:

$$E = \int_a^b f(x) dx - \frac{h}{3}[f(a) + 4f(\frac{a+b}{2}) + f(b)] = -\frac{h^5}{90}f^{(4)}(\xi) = -\frac{(b-a)^5}{2880}f^{(4)}(\xi)$$

for some $\xi \in (a, b)$.

Resulting order: The local error is $O(h^5)$. For composite Simpson's with n subintervals ($N = n/2$ application of the rule), the global error is:

$$|E_n| \leq \frac{(b-a)h^4}{180} \max_{x \in [a,b]} |f^{(4)}(x)| = O(h^4),$$

making composite Simpson's $O(h^4)$ globally – two orders better than the trapezoidal rule.

Question 76 — Romberg Integration

(a) Explain how Romberg integration is obtained by applying Richardson extrapolation to the composite trapezoidal rule.

Solution: The composite trapezoidal rule has the **Euler–Maclaurin expansion**:

$$T_h = I + c_2h^2 + c_4h^4 + c_6h^6 + \dots$$

where $I = \int_a^b f(x) dx$ and c_{2k} are constants depending on f and its derivatives at the endpoints.

The error is a power series in h^2 . This means we can apply Richardson extrapolation *repeatedly* to eliminate successive terms. Each level of extrapolation increases the order by 2.

Define $R_{k,1} = T_{h_k}$ where $h_k = (b-a)/2^{k-1}$ (halving the step each time). Then $R_{k,j}$ is the Richardson extrapolated estimate at level j , eliminating terms up to h^{2j-2} .

(b) Derive the recursion $R_{k,j} = R_{k,j-1} + \frac{R_{k,j-1} - R_{k-1,j-1}}{4^{j-1} - 1}$.

Solution: At level $j-1$, both $R_{k,j-1}$ and $R_{k-1,j-1}$ satisfy:

$$R_{k,j-1} = I + c_{2j} h_k^{2j-2} + O(h_k^{2j}), \quad R_{k-1,j-1} = I + c_{2j} h_{k-1}^{2j-2} + O(h_{k-1}^{2j}).$$

Since $h_{k-1} = 2h_k$, we have $h_{k-1}^{2j-2} = 4^{j-1} h_k^{2j-2}$.

Richardson extrapolation to cancel the h_k^{2j-2} term:

$$4^{j-1} R_{k,j-1} - R_{k-1,j-1} = (4^{j-1} - 1)I + O(h_k^{2j}).$$

Dividing by $4^{j-1} - 1$:

$$R_{k,j} = \frac{4^{j-1} R_{k,j-1} - R_{k-1,j-1}}{4^{j-1} - 1} = R_{k,j-1} + \frac{R_{k,j-1} - R_{k-1,j-1}}{4^{j-1} - 1}.$$

(c) Construct the Romberg table up to level $R_{3,3}$ for $I = \int_0^1 \ln(1+x) dx$.

Solution: Exact: $I = [(1+x) \ln(1+x) - x]_0^1 = 2 \ln 2 - 1 \approx 0.386294$.

Values: $\ln(1) = 0$, $\ln(1.25) \approx 0.223144$, $\ln(1.5) \approx 0.405465$, $\ln(1.75) \approx 0.559616$, $\ln(2) \approx 0.693147$.

$$R_{1,1} (h=1): \frac{1}{2} [\ln(1) + \ln(2)] = \frac{0.693147}{2} = 0.346574.$$

$$R_{2,1} (h=1/2): \frac{1}{4} [\ln(1) + 2\ln(1.5) + \ln(2)] = \frac{1}{4} [0 + 0.810930 + 0.693147] = \frac{1.504077}{4} = 0.376019.$$

$$R_{3,1} (h=1/4): \frac{1}{8} [0 + 2(0.223144) + 2(0.405465) + 2(0.559616) + 0.693147] = \frac{1}{8} [0.446288 + 0.810930 + 1.119232 + 0.693147] = \frac{3.069597}{8} = 0.383700.$$

$$R_{2,2}: R_{2,1} + \frac{R_{2,1} - R_{1,1}}{3} = 0.376019 + \frac{0.029445}{3} = 0.376019 + 0.009815 = 0.385834.$$

$$R_{3,2}: R_{3,1} + \frac{R_{3,1} - R_{2,1}}{3} = 0.383700 + \frac{0.007681}{3} = 0.383700 + 0.002560 = 0.386260.$$

$$R_{3,3}: R_{3,2} + \frac{R_{3,2} - R_{2,2}}{15} = 0.386260 + \frac{0.000426}{15} = 0.386260 + 0.000028 = \mathbf{0.386288}.$$

$k \backslash j$	1	2	3
1	0.346574		
2	0.376019	0.385834	
3	0.383700	0.386260	0.386288

Error of $R_{3,3}$: $|0.386294 - 0.386288| \approx 6 \times 10^{-6}$, far better than $R_{1,1}$ (error ≈ 0.040).

Question 77 — Gaussian Quadrature and Degree of Exactness

(a) Derive the 2-point Gauss–Legendre rule on $[-1, 1]$ by matching moments (solve for nodes and weights).

Solution: By symmetry, use nodes $\pm x_1$ and equal weights w . We need to integrate $1, x, x^2, x^3$ exactly.

$$\int_{-1}^1 1 dx = 2: \text{Rule gives } w \cdot 1 + w \cdot 1 = 2w. \text{ So } w = 1.$$

$$\int_{-1}^1 x dx = 0: \text{Rule gives } w(-x_1) + w(x_1) = 0. \text{ (automatic by symmetry).}$$

$$\int_{-1}^1 x^2 dx = 2/3: \text{Rule gives } wx_1^2 + wx_1^2 = 2x_1^2. \text{ So } 2x_1^2 = 2/3 \Rightarrow x_1 = 1/\sqrt{3}.$$

$\int_{-1}^1 x^3 dx = 0$: Automatic by symmetry.

Result: Nodes $\pm 1/\sqrt{3} \approx \pm 0.577350$, weights $w_1 = w_2 = 1$.

$$\int_{-1}^1 f(x) dx \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right).$$

(b) Prove it integrates all polynomials of degree ≤ 3 exactly.

Solution: A polynomial $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ is a linear combination of $1, x, x^2, x^3$.

Exact integrals: $\int_{-1}^1 1 = 2$, $\int_{-1}^1 x = 0$, $\int_{-1}^1 x^2 = 2/3$, $\int_{-1}^1 x^3 = 0$.

G-L rule: $p(-1/\sqrt{3}) + p(1/\sqrt{3})$.

For 1: gives $1 + 1 = 2$. For x : gives $-1/\sqrt{3} + 1/\sqrt{3} = 0$. For x^2 : gives $1/3 + 1/3 = 2/3$. For x^3 : gives $(-1/\sqrt{3})^3 + (1/\sqrt{3})^3 = -1/(3\sqrt{3}) + 1/(3\sqrt{3}) = 0$.

Since the rule integrates each basis polynomial exactly, it integrates all degree ≤ 3 polynomials exactly. $\square$

(c) Compare the degree of exactness of Gauss–Legendre rules to equally spaced Newton–Cotes rules as the number of points increases.

Solution:

- **n -point Gauss–Legendre:** Degree of exactness $= 2n - 1$. With n nodes, we have $2n$ free parameters (nodes and weights), allowing exact integration of polynomials up to degree $2n - 1$.
- **n -point Newton–Cotes (equally spaced):** Degree of exactness $= n - 1$ for n odd, n for n even.

Comparison: For $n = 2$: G-L gives degree 3, Trapezoid gives degree 1. For $n = 3$: G-L gives degree 5, Simpson’s gives degree 3. G-L achieves *double* the degree of exactness compared to Newton–Cotes for the same number of function evaluations. This is the key advantage of Gaussian quadrature for smooth integrands.

Question 78 — Design an Adaptive Quadrature Algorithm

(a) Describe an adaptive Simpson’s method: how you split intervals, how you estimate local error, and how you decide to stop refining.

Solution: Idea: Apply Simpson’s rule on $[a, b]$ and compare to the sum of Simpson’s rule on $[a, m]$ and $[m, b]$ where $m = (a + b)/2$.

Error estimate: By Richardson extrapolation for Simpson’s ($O(h^5)$ local error):

$$\text{error} \approx \frac{|S(a, b) - S(a, m) - S(m, b)|}{15}.$$

where $S(a, b)$ denotes the Simpson’s rule approximation on $[a, b]$.

Stopping criterion: If the estimated error on $[a, b]$ is $\leq$ the tolerance $\times (b - a)/(b_{\text{total}} - a_{\text{total}})$, accept $S(a, m) + S(m, b)$ as the local contribution. Otherwise, split further.

(b) Provide pseudocode (high level) including stopping criteria.

Solution:

```

if errorest < tol : return Sleft + Sright // accept(more accurate estimate) else : return AdaptiveSimpson(f, a, m, tol)
AdaptiveSimpson(f, m, b, tol/2)
function Simpson(f, a, b): return (b-a)/6 * (f(a) + 4*f((a+b)/2) + f(b))

```

Guard against infinite recursion with a max depth parameter.

(c) Discuss at least two failure modes and how to mitigate them.

Solution:

1. **Discontinuities:** If f has a jump discontinuity between two quadrature nodes, the error estimate may be misleadingly small. The method may return an incorrect result without warning. *Mitigation:* Detect potential discontinuities by checking if $|f(m) - (f(a) + f(b))/2|$ is unusually large; bracket and locate discontinuities before integration.
2. **Oscillatory functions:** For highly oscillatory f (e.g., $f(x) = \sin(1000x)$), the algorithm may refine excessively everywhere, leading to very deep recursion and high cost. *Mitigation:* Use specialized oscillatory quadrature (e.g., Filon's method, Levin method) designed for such integrands.
3. **Sharp spikes:** A narrow spike between nodes may be missed by coarse initial sampling. *Mitigation:* Use a finer initial grid to ensure spikes are captured before adaptive refinement begins.

Question 79 — Write an Optimization Problem in Standard Form

Consider: minimize $(x_1 - 2)^2 + (x_2 + 1)^2$ s.t. $1 \leq x_1 \leq 3$, $x_1 + x_2 = 1$, $x_2^2 \leq 4$.

(a) Rewrite in standard form with $f_i(x) \leq 0$ and $h_i(x) = 0$.

Solution: The box constraint $1 \leq x_1 \leq 3$ gives two inequalities. The inequality $x_2^2 \leq 4$ is already in the right form.

Inequality constraints $f_i(x) \leq 0$:

$$f_1(x) = 1 - x_1 \leq 0, \quad f_2(x) = x_1 - 3 \leq 0, \quad f_3(x) = x_2^2 - 4 \leq 0.$$

Equality constraint $h_1(x) = 0$:

$$h_1(x) = x_1 + x_2 - 1 = 0.$$

(b) Identify the functions f_i and h_i explicitly.

Solution:

$$f_1(x) = 1 - x_1, \quad f_2(x) = x_1 - 3, \quad f_3(x) = x_2^2 - 4, \quad h_1(x) = x_1 + x_2 - 1.$$

Note: f_1, f_2 are affine (convex), $f_3(x) = x_2^2 - 4$ is convex, h_1 is affine. The objective $(x_1 - 2)^2 + (x_2 + 1)^2$ is convex. So this is a **convex optimization problem**.

(c) Introduce slack variables to convert the inequalities into equalities with nonnegativity constraints.

Solution: Introduce $s_1, s_2, s_3 \geq 0$:

$$f_i(x) + s_i = 0, \quad s_i \geq 0.$$

Explicitly:

$$\begin{aligned} 1 - x_1 + s_1 &= 0 & (s_1 \geq 0) \\ x_1 - 3 + s_2 &= 0 & (s_2 \geq 0) \\ x_2^2 - 4 + s_3 &= 0 & (s_3 \geq 0) \\ x_1 + x_2 - 1 &= 0 & (\text{original equality}) \end{aligned}$$

The augmented problem in $(x_1, x_2, s_1, s_2, s_3)$ is equivalent to the original.

Question 80 — Convexity and Convex Reformulation

Consider: maximize $x_1^2 + x_2^2$ s.t. $\frac{x_1}{1+x_2} \leq 0$, $(x_1 + x_2)^2 = 0$.

(a) Explain why this is not a convex optimization problem.

Solution: Three convexity violations:

1. **Objective:** Maximizing $x_1^2 + x_2^2$ (a convex function) is *not* a convex optimization problem. Convex optimization requires *minimizing* a convex function.
2. **Inequality constraint:** The function $g(x) = x_1/(1+x_2^2)$ is not convex in general (it is a ratio whose second derivatives can be negative).
3. **Equality constraint:** $(x_1 + x_2)^2 = 0$ is a nonlinear (quadratic) equality, which violates the requirement that equality constraints be affine. (Though geometrically equivalent to $x_1 + x_2 = 0$, it is not written in affine form.)

(b) Replace constraints by equivalent ones to obtain a convex optimization problem with the same feasible set.

Solution: Feasible set analysis: $\frac{x_1}{1+x_2^2} \leq 0 \iff x_1 \leq 0$ (since $1+x_2^2 > 0$ always). $(x_1 + x_2)^2 = 0 \iff x_1 + x_2 = 0$ (the only real root).

Equivalent constraints:

$$f_1(x) = x_1 \leq 0 \quad (\text{convex/affine}), \quad h_1(x) = x_1 + x_2 = 0 \quad (\text{affine}).$$

Both are now in convex standard form.

(c) Rewrite the final problem explicitly and justify convexity.

Solution: Problem as reformulated:

$$\text{maximize } x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 \leq 0, \quad x_1 + x_2 = 0.$$

The constraints are now convex (affine). However, note that maximizing $x_1^2 + x_2^2$ is still non-convex as an optimization problem. To fully obtain a convex problem, restate as:

$$\text{minimize } -(x_1^2 + x_2^2) \quad \text{s.t.} \quad x_1 \leq 0, \quad x_1 + x_2 = 0.$$

The objective $-(x_1^2 + x_2^2)$ is concave (not convex), so this is actually a **concave minimization** – in general still non-convex. However, on the feasible set (a ray $\{(-t, t) : t \geq 0\}$), the problem is unbounded above (as $t \rightarrow \infty$, $x_1^2 + x_2^2 = 2t^2 \rightarrow \infty$), confirming no finite optimum exists for the original maximization.

Question 81 — Convexity Proof and Optimization

Let $f(x) = x \log x$ with domain $x > 0$.

(a) Prove f is convex on $(0, \infty)$.

Solution: Compute the second derivative:

$$f'(x) = \log x + 1, \quad f''(x) = \frac{1}{x}.$$

Since $x > 0$, we have $f''(x) = 1/x > 0$ for all $x \in (0, \infty)$. A function with positive second derivative on an interval is strictly convex on that interval. $\square$

(b) Find the global minimizer of f on $(0, \infty)$ and compute the minimum value.

Solution: Since f is convex and differentiable, the global minimizer satisfies $f'(x^*) = 0$:

$$\log x^* + 1 = 0 \implies \log x^* = -1 \implies x^* = e^{-1} = \frac{1}{e}.$$

Minimum value:

$$f(x^*) = \frac{1}{e} \log \left(\frac{1}{e} \right) = \frac{1}{e} \cdot (-1) = -\frac{1}{e} \approx -0.3679.$$

(c) Discuss whether the minimum is attained and what happens near the boundary $x \downarrow 0$.

Solution: Minimum is attained: Yes, $x^* = 1/e \in (0, \infty)$, which is in the interior of the domain. The minimum value $-1/e$ is achieved.

Near the boundary $x \downarrow 0$:

$$\lim_{x \rightarrow 0^+} x \log x = \lim_{x \rightarrow 0^+} \frac{\log x}{1/x} \stackrel{L'H}{=} \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0.$$

As $x \rightarrow 0^+$, $f(x) \rightarrow 0$. Since $f(x^*) = -1/e < 0 < f(x \rightarrow 0^+)$, the minimum is indeed in the interior. The function rises back toward 0 as we approach the left boundary.

Question 82 — Newton's Method for Minimization

(a) Derive the Newton step for unconstrained minimization from the second-order Taylor approximation of f .

Solution: Approximate f near x_k by its second-order Taylor expansion:

$$\hat{f}(x_k + d) \approx f(x_k) + \nabla f(x_k)^T d + \frac{1}{2} d^T \nabla^2 f(x_k) d.$$

Minimize $\hat{f}$ over d by setting the gradient to zero:

$$\nabla_d \hat{f} = \nabla f(x_k) + \nabla^2 f(x_k) d = 0.$$

Solving: $d^* = -[\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$ (Newton direction).

The Newton update is $x_{k+1} = x_k + d^* = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$.

(b) State conditions under which the Newton direction is a descent direction.

Solution: The Newton direction $d = -H^{-1}g$ (where $H = \nabla^2 f(x_k)$, $g = \nabla f(x_k)$) is a **descent direction** if $g^T d < 0$:

$$g^T d = g^T (-H^{-1}g) = -g^T H^{-1}g < 0.$$

This holds if and only if $H = \nabla^2 f(x_k)$ is **positive definite**. If the Hessian is indefinite or negative definite, the Newton direction may not be a descent direction.

(c) For $f(x) = \cos x + x^2/10$, write the Newton iteration formula explicitly.

Solution: $f'(x) = -\sin x + x/5$, $f''(x) = -\cos x + 1/5$.

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)} = x_k - \frac{-\sin x_k + x_k/5}{-\cos x_k + 1/5}.$$

(d) Starting from $x_0 = 1$, compute or approximate three Newton iterates and comment on convergence.

Solution: $f'(1) = -\sin(1) + 0.2 \approx -0.8415 + 0.2 = -0.6415$. $f''(1) = -\cos(1) + 0.2 \approx -0.5403 + 0.2 = -0.3403$.

Note: $f''(1) < 0$! The Hessian is *negative*, meaning $x_0 = 1$ is near a local *maximum* of f .

$x_1 = 1 - (-0.6415)/(-0.3403) = 1 - 1.885 \approx -0.885$.

At $x_1 \approx -0.885$: $f''(-0.885) = -\cos(-0.885) + 0.2 \approx -0.637 + 0.2 = -0.437 < 0$ still negative.

The method diverges/oscillates because $f''(x) = -\cos x + 0.2$; since $|\cos x| \leq 1$ and $0.2 < 1$, $f''(x) < 0$ for many x (near $x = 0$). Newton's method for minimization fails without damping here.

(e) Explain why damping/line search is often needed in practice.

Solution: Without damping:

- If the Hessian is **indefinite**, the Newton step may be uphill.
- Near saddle points or far from a minimum, the quadratic model can be a poor approximation and Newton steps can overshoot badly.
- The pure Newton step has no guarantee of decreasing f .

With a line search (e.g., Armijo backtracking), we ensure $f(x_{k+1}) < f(x_k)$ at each step while still benefiting from Newton's fast local convergence. Modified Newton methods (e.g., adding a multiple of I to make the Hessian positive definite) also help.

Question 83 — Dual Function Properties

(a) Prove that for any $\lambda \geq 0$ and any ν , $g(\lambda, \nu) \leq p^*$.

Solution: Let x be any primal feasible point: $f_i(x) \leq 0$ for all i , and $h_j(x) = 0$ for all j .

Since $\lambda \geq 0$ and $f_i(x) \leq 0$: $\lambda_i f_i(x) \leq 0$ for each i . Since $h_j(x) = 0$: $\nu_j h_j(x) = 0$.

Therefore:

$$L(x, \lambda, \nu) = f_0(x) + \sum_i \lambda_i f_i(x) + \sum_j \nu_j h_j(x) \leq f_0(x).$$

By definition, $g(\lambda, \nu) = \inf_{x'} L(x', \lambda, \nu) \leq L(x, \lambda, \nu) \leq f_0(x)$.

This holds for *every* primal feasible x , so:

$$g(\lambda, \nu) \leq \inf_{x \text{ feasible}} f_0(x) = p^*. \quad \square$$

(b) Prove that $g(\lambda, \nu)$ is concave in (λ, ν) .

Solution: For fixed x , the Lagrangian $L(x, \lambda, \nu) = f_0(x) + \sum_i \lambda_i f_i(x) + \sum_j \nu_j h_j(x)$ is an **affine** (hence concave) function of (λ, ν) .

The dual function $g(\lambda, \nu) = \inf_x L(x, \lambda, \nu)$ is a **pointwise infimum** of a family of affine (concave) functions of (λ, ν) .

Fact: The pointwise infimum of any family of concave functions is concave:

$$\inf_x [\alpha L(x, \lambda, \nu) + (1 - \alpha)L(x, \mu, \xi)] \geq \alpha \inf_x L(x, \lambda, \nu) + (1 - \alpha) \inf_x L(x, \mu, \xi).$$

Wait, more precisely: for any $\alpha \in [0, 1]$,

$$\begin{aligned} g(\alpha(\lambda, \nu) + (1 - \alpha)(\mu, \xi)) &= \inf_x L(x, \alpha\lambda + (1 - \alpha)\mu, \alpha\nu + (1 - \alpha)\xi) \\ &= \inf_x [\alpha L(x, \lambda, \nu) + (1 - \alpha)L(x, \mu, \xi)] \geq \alpha \inf_x L(x, \lambda, \nu) + (1 - \alpha) \inf_x L(x, \mu, \xi) = \alpha g(\lambda, \nu) + (1 - \alpha)g(\mu, \xi). \end{aligned}$$

Hence g is concave. $\square$

(c) Define the Lagrange dual problem and dual feasibility.

Solution: Lagrange dual problem: Maximize the dual function over dual feasible points:

$$\text{maximize } g(\lambda, \nu) \quad \text{subject to } \lambda \geq 0.$$

(No sign restriction on ν .) The dual optimal value is $d^* = \sup_{\lambda \geq 0, \nu} g(\lambda, \nu)$.

Dual feasibility: A point (λ, ν) is dual feasible if $\lambda \geq 0$ and $g(\lambda, \nu) > -\infty$.

(d) Explain how a dual feasible point serves as a certificate (lower bound) for the primal optimum.

Solution: By part (a), any dual feasible (λ, ν) satisfies $g(\lambda, \nu) \leq p^*$. Therefore, if we find (λ, ν) with $\lambda \geq 0$ and compute $g(\lambda, \nu) = L_0$ for some value L_0 , we know:

$$p^* \geq L_0.$$

This provides a **certified lower bound**: no feasible point can have objective value below L_0 . If we also have a primal feasible point x with $f_0(x) = U_0$, then $L_0 \leq p^* \leq U_0$, giving a guaranteed gap $U_0 - L_0$.

Question 84 — Deriving the Dual Using Conjugates

Let f_0 be closed, proper, convex. Minimize $f_0(x)$ s.t. $Ax \leq b$, $Cx = d$.

(a) Derive the dual function and show it can be written as $g(\lambda, \nu) = -b^T \lambda - d^T \nu - f_0^*(-A^T \lambda - C^T \nu)$.

Solution: Lagrangian: $L(x, \lambda, \nu) = f_0(x) + \lambda^T (Ax - b) + \nu^T (Cx - d)$, $\lambda \geq 0$.

$$= f_0(x) + (A^T \lambda + C^T \nu)^T x - b^T \lambda - d^T \nu.$$

Dual function:

$$g(\lambda, \nu) = \inf_x [f_0(x) + (A^T \lambda + C^T \nu)^T x] - b^T \lambda - d^T \nu.$$

Using the definition of conjugate $f_0^*(y) = \sup_x [y^T x - f_0(x)]$:

$$\inf_x [f_0(x) + y^T x] = -\sup_x [-y^T x - f_0(x)] = -f_0^*(-y).$$

With $y = A^T\lambda + C^T\nu$:

$$g(\lambda, \nu) = -b^T\lambda - d^T\nu - f_0^*(-A^T\lambda - C^T\nu).$$

(b) State the domain of (λ, ν) for which $g(\lambda, \nu) > -\infty$.

Solution: $g(\lambda, \nu) > -\infty$ if and only if $-A^T\lambda - C^T\nu \in \text{dom}(f_0^*)$, i.e., the conjugate is finite at $-A^T\lambda - C^T\nu$. The domain of the dual function (where it is finite) is $\{(\lambda, \nu) : -A^T\lambda - C^T\nu \in \text{dom}(f_0^*)\}$.

(c) Interpret the dual as a conjugate-based dual (Fenchel-type dual) in words.

Solution: The dual function penalizes the constraint violation through the Lagrange multipliers (λ, ν) , and the resulting optimization over x produces the conjugate f_0^* . The dual problem is: find (λ, ν) with $\lambda \geq 0$ to maximize $-b^T\lambda - d^T\nu - f_0^*(-A^T\lambda - C^T\nu)$. This is the **Fenchel dual**: it expresses the dual in terms of the conjugate of f_0 , showing that the dual of an LP/convex program with linear constraints naturally involves the conjugate of the objective.

Question 85 — Prove Weak Duality

Provide a complete proof of weak duality: for a minimization problem in standard form, show $d^* \leq p^*$.

Solution: Setup: Consider the problem: minimize $f_0(x)$ s.t. $f_i(x) \leq 0$ ($i = 1, \dots, m$), $h_j(x) = 0$ ($j = 1, \dots, p$).

Step 1: Take any primal feasible point x (so $f_i(x) \leq 0$ and $h_j(x) = 0$) and any dual feasible point (λ, ν) (so $\lambda \geq 0$).

Step 2: Bound the Lagrangian:

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \underbrace{\lambda_i}_{\geq 0} \underbrace{f_i(x)}_{\leq 0} + \sum_{j=1}^p \nu_j \underbrace{h_j(x)}_{=0} \leq f_0(x).$$

Step 3: The dual function satisfies:

$$g(\lambda, \nu) = \inf_{x'} L(x', \lambda, \nu) \leq L(x, \lambda, \nu) \leq f_0(x).$$

Step 4: Since Step 3 holds for *every* primal feasible x :

$$g(\lambda, \nu) \leq \inf_{\text{primal feasible}} f_0(x) = p^*.$$

Step 5: Since Step 4 holds for every dual feasible (λ, ν) :

$$d^* = \sup_{\lambda \geq 0, \nu} g(\lambda, \nu) \leq p^*. \quad \square$$

Question 86 — Prove Complementary Slackness

Assume strong duality holds and primal/dual optimal solutions exist. Prove $\lambda_i^* f_i(x^*) = 0$ for $i = 1, \dots, m$, and show x^* minimizes $L(x, \lambda^*, \nu^*)$ over x .

Solution: **Setup:** $p^* = d^*$, x^* is primal optimal, (λ^*, ν^*) is dual optimal.

Chain of inequalities:

$$p^* = f_0(x^*) \stackrel{(1)}{\geq} L(x^*, \lambda^*, \nu^*) \stackrel{(2)}{\geq} g(\lambda^*, \nu^*) = d^* = p^*.$$

(1) From weak duality proof: $L(x, \lambda, \nu) \leq f_0(x)$ for primal feasible x and dual feasible (λ, ν) .

(2) $g(\lambda^*, \nu^*) = \inf_x L(x, \lambda^*, \nu^*) \leq L(x^*, \lambda^*, \nu^*)$.

Since $p^* \geq L(x^*, \lambda^*, \nu^*) \geq g(\lambda^*, \nu^*) = p^*$, both inequalities must be equalities.

From equality (2): $g(\lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*)$, meaning x^* **minimizes** $L(x, \lambda^*, \nu^*)$ over x .

From equality (1):

$$f_0(x^*) = L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \sum_i \lambda_i^* f_i(x^*).$$

Therefore $\sum_i \lambda_i^* f_i(x^*) = 0$.

Since each term $\lambda_i^* f_i(x^*) \leq 0$ (as $\lambda_i^* \geq 0$ and $f_i(x^*) \leq 0$), and their sum is zero, each term must be zero:

$$\lambda_i^* f_i(x^*) = 0 \quad \text{for all } i. \quad \square$$

Question 87 — KKT Conditions: Necessity and Sufficiency

(a) Derive the KKT conditions for a differentiable problem in standard form.

Solution: At optimality, if the problem is convex and Slater's holds (or a CQ holds generally), the KKT conditions must be satisfied by x^* and some (λ^*, ν^*) :

(1) **Primal feasibility:** $f_i(x^*) \leq 0$, $h_j(x^*) = 0$.

(2) **Dual feasibility:** $\lambda_i^* \geq 0$.

(3) **Complementary slackness:** $\lambda_i^* f_i(x^*) = 0$.

(4) **Stationarity:** $\nabla f_0(x^*) + \sum_i \lambda_i^* \nabla f_i(x^*) + \sum_j \nu_j^* \nabla h_j(x^*) = 0$.

The stationarity condition comes from x^* minimizing $L(x, \lambda^*, \nu^*)$ (by Q86), which for a differentiable unconstrained minimization requires zero gradient.

(b) Under convexity and Slater's condition, prove that KKT conditions are sufficient for optimality.

Solution: Suppose (x^*, λ^*, ν^*) satisfy KKT. We show x^* is globally optimal.

Key: From stationarity, x^* minimizes $L(x, \lambda^*, \nu^*)$ over all x (since L is convex in x by convexity of f_0, f_i , and the gradient condition holds).

Therefore $g(\lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*)$.

From complementary slackness and primal/dual feasibility:

$$g(\lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \underbrace{\sum_i \lambda_i^* f_i(x^*)}_{=0} + \underbrace{\sum_j \nu_j^* h_j(x^*)}_{=0} = f_0(x^*).$$

By weak duality: $f_0(x^*) = g(\lambda^*, \nu^*) \leq p^* \leq f_0(x^*)$. Hence $f_0(x^*) = p^*$, so x^* is globally optimal. $\square$

(c) Explain which parts fail (or become only necessary) when the problem is nonconvex.

Solution: For nonconvex problems:

- **Necessity:** KKT conditions remain necessary for a local minimum (under constraint qualifications like LICQ or MFCQ), but prove only local, not global optimality.
- **Sufficiency fails:** A KKT point may be a local minimum, local maximum, or saddle point. The stationarity condition $\nabla_x L = 0$ does not guarantee x^* globally minimizes $L(\cdot, \lambda^*, \nu^*)$ if L is non-convex.
- **Strong duality may fail:** There may be a nonzero duality gap ($d^* < p^*$), as seen in Q89.

Question 88 — Strong Duality: Equality Constrained Least Squares

Consider: minimize $x^T x$ subject to $Ax = b$, where $A \in \mathbb{R}^{p \times n}$.

(a) Write the Lagrangian and dual function.

Solution: Lagrangian: $L(x, \nu) = x^T x + \nu^T (Ax - b) = \|x\|_2^2 + \nu^T Ax - \nu^T b$.

Dual function: Minimize over x (unconstrained convex quadratic): $\nabla_x L = 2x + A^T \nu = 0 \Rightarrow x^* = -A^T \nu / 2$.

$$g(\nu) = \left(-\frac{A^T \nu}{2}\right)^T \left(-\frac{A^T \nu}{2}\right) + \nu^T A \left(-\frac{A^T \nu}{2}\right) - \nu^T b = \frac{\nu^T A A^T \nu}{4} - \frac{\nu^T A A^T \nu}{2} - \nu^T b = -\frac{\nu^T A A^T \nu}{4} - \nu^T b.$$

(b) Write the dual problem explicitly.

Solution:

$$\text{maximize } g(\nu) = -\frac{1}{4} \nu^T (A A^T) \nu - \nu^T b \quad \text{over } \nu \in \mathbb{R}^p.$$

This is an unconstrained concave quadratic (no sign constraint on ν since the equality constraint has no sign restriction). Taking the gradient and setting to zero: $-\frac{1}{2} A A^T \nu^* - b = 0 \Rightarrow A A^T \nu^* = -2b$. If $A A^T$ is invertible: $\nu^* = -2(A A^T)^{-1} b$.

(c) Show that strong duality holds whenever the primal is feasible.

Solution: The primal problem is minimizing a strictly convex objective ($\|x\|^2$) over the affine set $\{Ax = b\}$. If the primal is feasible ($b \in \text{Range}(A)$), the minimum exists uniquely. Slater's condition for an equality-constrained problem just requires feasibility. Since the objective is strictly convex and the constraint is affine, strong duality holds automatically (the dual gap is zero). Formally, since $L(x, \nu)$ is convex in x and the infimum is attained (the quadratic is strictly convex), the dual function is finite everywhere and strong duality follows.

(d) If $b \in \text{Range}(A)$, express the primal optimizer x^* in terms of A and b .

Solution: From $x^* = -A^T \nu^* / 2$ and $\nu^* = -2(AA^T)^{-1}b$:

$$x^* = -A^T \cdot \frac{-2(AA^T)^{-1}b}{2} = A^T(AA^T)^{-1}b.$$

This is the **Moore–Penrose pseudoinverse**: $x^* = A^\dagger b = A^T(AA^T)^{-1}b$ (for full row-rank A). It gives the *minimum-norm* solution to $Ax = b$.

Question 89 — Strong Duality Does Not Hold (Nonconvex)

Consider: minimize x^2 subject to $x^3 = 1$.

(a) Find the primal optimizer and optimal value p^* .

Solution: The constraint $x^3 = 1$ has the unique real solution $x = 1$.

$$x^* = 1, \quad p^* = (x^*)^2 = 1.$$

(b) Form the Lagrangian and compute the dual function $g(\nu) = \inf_x L(x, \nu)$.

Solution: Lagrangian: $L(x, \nu) = x^2 + \nu(x^3 - 1)$.

Dual function: $g(\nu) = \inf_{x \in \mathbb{R}} [x^2 + \nu x^3 - \nu]$.

For $\nu > 0$: as $x \rightarrow -\infty$, $\nu x^3 \rightarrow -\infty$, so $g(\nu) = -\infty$. For $\nu < 0$: as $x \rightarrow +\infty$, $\nu x^3 \rightarrow -\infty$, so $g(\nu) = -\infty$. For $\nu = 0$: $g(0) = \inf_x x^2 = 0$.

$$g(\nu) = \begin{cases} 0 & \nu = 0 \\ -\infty & \nu \neq 0. \end{cases}$$

(c) Solve the dual problem and obtain d^* .

Solution: Dual: maximize $g(\nu)$ over $\nu \in \mathbb{R}$. The only finite value is $g(0) = 0$, so $d^* = 0$.

(d) Compute the duality gap $p^* - d^*$ and conclude that strong duality fails.

Solution:

$$p^* - d^* = 1 - 0 = 1 > 0.$$

The duality gap is 1. Strong duality **fails** because the problem is nonconvex (the equality constraint $x^3 = 1$ is not affine, and the feasible set is non-convex: it is just one point $\{1\}$ in the nonconvex set defined by a cubic).

Question 90 — Recover Primal Optimum Given Dual Optimum

LP: minimize $2x_1 + x_2$ s.t. $x_1 + x_2 \geq 2$, $x_1, x_2 \geq 0$.

(a) Write the dual LP.

Solution: Primal: $\min c^T x$, $Ax \geq b$, $x \geq 0$, where $c = [2, 1]^T$, $A = [1, 1]$, $b = 2$.

Dual LP: $\max b^T y$, $A^T y \leq c$, $y \geq 0$:

$$\text{maximize } 2y \quad \text{s.t.} \quad y \leq 2, \quad y \leq 1, \quad y \geq 0.$$

i.e., $\max 2y$ s.t. $0 \leq y \leq 1$.

(b) Suppose the dual optimal multiplier is $y^* = 1$. Use complementary slackness to recover a primal optimal solution.

Solution: Complementary slackness conditions:

- **Primal:** $y^*(x_1^* + x_2^* - 2) = 0$. Since $y^* = 1 \neq 0$: $x_1^* + x_2^* = 2$.
- **Dual for x_1 :** $x_1^*(c_1 - y^* \cdot 1) = x_1^*(2 - 1) = x_1^* = 0$.
- **Dual for x_2 :** $x_2^*(c_2 - y^* \cdot 1) = x_2^*(1 - 1) = 0$ (satisfied for any x_2^*).

From $x_1^* = 0$ and $x_1^* + x_2^* = 2$: $x_2^* = 2$.

$$\boxed{x_1^* = 0, \quad x_2^* = 2.}$$

(c) Verify primal feasibility, dual feasibility, and equality of primal/dual objective values at optimum.

Solution: **Primal feasibility:** $x_1^* + x_2^* = 0 + 2 = 2 \geq 2$, $x_1^* = 0 \geq 0$, $x_2^* = 2 \geq 0$.

Dual feasibility: $y^* = 1 \geq 0$, $y^* = 1 \leq 2$, $y^* = 1 \leq 1$.

Objective equality: Primal: $2(0) + 1(2) = 2$. Dual: $2 \cdot 1 = 2$. Equal. Strong duality holds.

Question 91 — Grid World as an MDP

4×4 grid, terminal at $(4, 4)$, start at $(1, 1)$, actions {Up, Down, Left, Right}, reward $-1/\text{step}$, 0 at terminal.

(a) Specify S , A , the transition rule, and the reward function.

Solution: $S = \{(i, j) : 1 \leq i, j \leq 4\} \cup \{T\}$, where $T = (4, 4)$ is the terminal state ($|S| = 16$).

$A = \{\text{Up, Down, Left, Right}\}$.

Transition: Deterministic. From state (i, j) , action Up $\rightarrow (i - 1, j)$ if $i > 1$, else stays at (i, j) (wall). Similarly for other directions. From terminal: stays at T .

Reward: $r(s, a) = -1$ for all non-terminal s ; $r(T, a) = 0$.

(b) Choose a discount factor γ and justify your choice for an episodic task.

Solution: For an episodic task (guaranteed to terminate), $\gamma = 1$ is appropriate and simplifies analysis: G_t is just the negative of the number of steps to terminal, which is what we want to minimize.

Alternatively, $\gamma = 0.99$ (close to 1) can be used if we want the algorithm to prefer shorter paths without worrying about convergence of the undiscounted sum.

Justification: Since every episode terminates (finite grid, no loops if agent moves toward terminal), the return $G_t = \sum_{k=0}^T -1 \cdot (-1)^k$ is always finite even with $\gamma = 1$.

(c) Write the Bellman optimality equations for $V^*(s)$.

Solution:

$$V^*(s) = \max_{a \in A} [r(s, a) + \gamma V^*(s'(s, a))] \quad \text{for } s \neq T,$$

$$V^*(T) = 0.$$

where $s'(s, a)$ is the next state (deterministic). The optimal policy is $\pi^*(s) = \arg \max_a [r(s, a) + \gamma V^*(s'(s, a))]$.

(d) Describe a simple indexing scheme to map (i, j) grid locations into integer state IDs.

Solution: $\text{ID}(i, j) = (i - 1) \cdot 4 + (j - 1)$, giving IDs $0, 1, \dots, 15$ for $(i, j) \in \{1, \dots, 4\}^2$.

For example: $(1, 1) \rightarrow 0$, $(1, 4) \rightarrow 3$, $(4, 1) \rightarrow 12$, $(4, 4) \rightarrow 15$ (terminal).

Inverse: $i = \lfloor \text{ID}/4 \rfloor + 1$, $j = (\text{ID} \bmod 4) + 1$.

Question 92 — Bellman Equations and Q-learning

(a) Derive the Bellman expectation equation for $Q^\pi(s, a)$.

Solution:

$$\begin{aligned}
 Q^\pi(s, a) &= \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] \\
 &= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a] \\
 &= \sum_{s'} P(s' \mid s, a) [r(s, a, s') + \gamma \mathbb{E}_\pi[G_{t+1} \mid S_{t+1} = s']] \\
 &= \sum_{s'} P(s' \mid s, a) [r(s, a, s') + \gamma V^\pi(s')] \\
 &= \sum_{s'} P(s' \mid s, a) \left[r(s, a, s') + \gamma \sum_{a'} \pi(a' \mid s') Q^\pi(s', a') \right].
 \end{aligned}$$

(b) Derive the Bellman optimality equation for $Q^*(s, a)$.

Solution: The optimal Q-function satisfies:

$$Q^*(s, a) = \sum_{s'} P(s' \mid s, a) \left[r(s, a, s') + \gamma \max_{a'} Q^*(s', a') \right].$$

This follows from $V^*(s') = \max_{a'} Q^*(s', a')$ (the optimal value is achieved by taking the best action).

(c) Explain why Q-learning uses the update target $r + \gamma \max_{a'} Q(s', a')$.

Solution: Q-learning aims to learn Q^* , which satisfies the Bellman optimality equation. The TD target $r + \gamma \max_{a'} Q(s', a')$ is the right-hand side of this equation (with current Q as an approximation of Q^*). The Q-learning update moves the current estimate $Q(s, a)$ toward this target, iteratively converging to the fixed point Q^* .

(d) Explain why the Bellman optimality operator is a contraction mapping when $\gamma < 1$, and how this supports convergence.

Solution: Define the Bellman optimality operator $(TQ)(s, a) = \sum_{s'} P(s' \mid s, a) [r(s, a, s') + \gamma \max_{a'} Q(s', a')]$.

For any Q, Q' :

$$|(TQ)(s, a) - (TQ')(s, a)| = \gamma \left| \sum_{s'} P(s' \mid s, a) [\max_{a'} Q(s', a') - \max_{a'} Q'(s', a')] \right| \leq \gamma \max_{s'} \max_{a'} |Q(s', a') - Q'(s', a')|.$$

Taking the sup over (s, a) : $\|TQ - TQ'\|_\infty \leq \gamma \|Q - Q'\|_\infty$.

Since $\gamma < 1$, T is a **contraction** with factor γ . By the **Banach fixed-point theorem**, T has a unique fixed point Q^* , and repeated application of T starting from any Q_0 converges to Q^* .

Question 93 — Q-learning on Grid World Without Experience Replay

Using the grid world from Q91, $Q_0(s, a) = 0$, $\alpha = 0.5$, $\gamma = 0.9$. Episode: $(1, 1) \xrightarrow{R, -1} (1, 2) \xrightarrow{R, -1} (1, 3) \xrightarrow{D, -1} (2, 3) \xrightarrow{D, -1} (3, 3)$.

(a) Perform the Q-learning update for each transition.

Solution: Q-update: $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$.

Transition 1: $s = (1, 1)$, $a = \text{Right}$, $r = -1$, $s' = (1, 2)$. $\max_{a'} Q((1, 2), a') = 0$. $Q((1, 1), R) \leftarrow 0 + 0.5[-1 + 0.9(0) - 0] = -0.5$.

Transition 2: $s = (1, 2)$, $a = \text{Right}$, $r = -1$, $s' = (1, 3)$. $\max_{a'} Q((1, 3), a') = 0$. $Q((1, 2), R) \leftarrow 0 + 0.5[-1 + 0] = -0.5$.

Transition 3: $s = (1, 3)$, $a = \text{Down}$, $r = -1$, $s' = (2, 3)$. $\max_{a'} Q((2, 3), a') = 0$. $Q((1, 3), D) \leftarrow 0 + 0.5[-1 + 0] = -0.5$.

Transition 4: $s = (2, 3)$, $a = \text{Down}$, $r = -1$, $s' = (3, 3)$. $\max_{a'} Q((3, 3), a') = 0$. $Q((2, 3), D) \leftarrow 0 + 0.5[-1 + 0] = -0.5$.

(b) After these updates, which action would be greedy at state $(1, 1)$?

Solution: After the updates: $Q((1, 1), \text{Right}) = -0.5$; all other actions at $(1, 1)$ have $Q = 0$.

The greedy action maximizes $Q((1, 1), \cdot)$. Since $0 > -0.5$, any action *other than Right* (Up, Down, Left) is greedy (they all have $Q = 0$).

This is misleading – the agent hasn’t explored those directions yet, so $Q = 0$ doesn’t reflect their true value. This illustrates the failure of pure greedy selection.

(c) Explain how ε -greedy exploration would alter learning dynamics and why “always greedy” can fail early in training.

Solution: ε -greedy: With probability ε , take a random action; with probability $1 - \varepsilon$, take the greedy action. This ensures all state-action pairs are explored.

Why always greedy fails: Early in training, Q-values are initialized to 0 (or arbitrary values) and do not reflect reality. A greedy policy would avoid actions with any negative Q-value (like Right above), even if Right is actually the optimal direction. The agent gets stuck in a loop, never discovering the correct path to the goal. Exploration ($\varepsilon > 0$) ensures the agent eventually visits all states and builds correct Q-estimates.

Question 94 — Experience Replay

(a) Define experience replay (what is stored, how batches are sampled).

Solution: **Experience replay buffer $\mathcal{D}$:** A finite-capacity circular buffer storing tuples (s_t, a_t, r_t, s_{t+1}) – the agent’s past experiences.

Sampling: At each training step, a random minibatch $\{(s_i, a_i, r_i, s'_i)\}_{i=1}^B$ is sampled uniformly from $\mathcal{D}$. The Q-network is updated using gradient descent on the TD loss for this batch.

(b) Explain why experience replay improves sample efficiency and reduces temporal correlations.

Solution:

1. **Sample efficiency:** Each transition (s, a, r, s') is stored and can be reused in multiple gradient updates. In contrast, online learning discards each experience after one use. This is especially valuable when environment interactions are expensive.
2. **Reduces correlations:** Online RL samples consecutive transitions $(s_t, a_t, r_t, s_{t+1}), (s_{t+1}, a_{t+1}, r_{t+1}, s_{t+2})$, which are highly correlated (nearby in time and state space). Random minibatch sampling from the buffer decorrelates the training data, making gradient estimates more like i.i.d. samples, which is what SGD convergence theory requires.

(c) Explain what can go wrong if the replay buffer contains very stale data under a changing policy.

Solution: Q-learning is off-policy (can use transitions from old policies), but if the policy changes significantly, very old transitions may come from a distribution far from the current policy's distribution. The TD targets $r + \gamma \max_{a'} Q(s', a')$ computed for old transitions use the current Q -function but the transitions were generated by a different (possibly much worse) policy, leading to:

- **Biased gradient updates** that may not reflect the current policy's behavior.
- **Slower convergence** or divergence if the policy distribution shift is too large.

Mitigation: Use a prioritized replay buffer or limit the buffer size so old experiences expire.

(d) Give one reason experience replay may be unnecessary in small tabular problems, and one reason it can still be beneficial.

Solution: Unnecessary: In small tabular MDPs, every state-action pair can be visited frequently enough online, and each update is cheap ($O(1)$). The overhead of maintaining and sampling a buffer outweighs the benefit.

Still beneficial: Even in small problems, reducing temporal correlations can improve convergence speed. If the environment has rare but important transitions (e.g., receiving a large reward), storing and replaying them more often (prioritized replay) can accelerate learning.

Question 95 — Q-values and Value Function

(a) Define $V^\pi(s)$ and $Q^\pi(s, a)$ precisely.

Solution: Let $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$ be the discounted return from time t .

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s\right].$$

$$Q^\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, A_t = a\right].$$

$V^\pi(s)$ is the expected return from state s following π . $Q^\pi(s, a)$ is the expected return after taking action a in state s then following π .

(b) Prove that $V^\pi(s) = \sum_a \pi(a \mid s) Q^\pi(s, a)$.

Solution:

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_\pi[G_t \mid S_t = s] \\ &= \sum_a P(A_t = a \mid S_t = s) \cdot \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] \quad (\text{law of total expectation}) \\ &= \sum_a \pi(a \mid s) \cdot Q^\pi(s, a). \quad \square \end{aligned}$$

(c) For a deterministic policy $\pi(s)$, simplify the relationship between V^π and Q^π .

Solution: For a deterministic policy, $\pi(a \mid s) = \mathbf{1}[a = \pi(s)]$ (probability 1 on action $\pi(s)$, 0 on others). Then:

$$V^\pi(s) = \sum_a \pi(a \mid s) Q^\pi(s, a) = Q^\pi(s, \pi(s)).$$

The value equals the Q-value of the action chosen by the policy.

(d) Explain how a greedy policy improvement step uses Q^π .

Solution: Given the current policy π and its Q-function Q^π , define the **improved policy**:

$$\pi'(s) = \arg \max_a Q^\pi(s, a).$$

The **Policy Improvement Theorem** guarantees $V^{\pi'}(s) \geq V^\pi(s)$ for all s . Intuitively, always taking the best action according to Q^π is at least as good as following π . Repeated policy improvement (policy iteration) converges to π^* .

Question 96 — State and Action Sets for Chess, CartPole, and Tetris

(a) Chess:

Solution: **State:** Position of all 32 pieces on the 8×8 board (or a $8 \times 8 \times 12$ bitboard, one channel per piece type per player), plus side to move, castling rights, en passant square. **Discrete.**

Actions: All legal moves from the current position (~ 20 – 35 on average, up to 218 in extreme cases). **Discrete.**

Reward: $+1$ for win, -1 for loss, 0 for draw.

(b) CartPole:

Solution: **State:** $(x, \dot{x}, \theta, \dot{\theta})$ – cart position, cart velocity, pole angle, pole angular velocity. $\in \mathbb{R}^4$. **Continuous.**

Actions: Push cart left or right. **Discrete** ($|A| = 2$).

Reward: $+1$ for every timestep the pole remains upright (balancing task).

(c) Tetris (brick game):

Solution: **State:** Current grid configuration (20×10 binary matrix) + current falling piece type and orientation + next piece preview. **Discrete.**

Actions: Rotate piece ($0/90/180/270^\circ$) and translate horizontally; place/drop. **Discrete.**

Reward: Lines cleared per step (or $+1$ per line, large bonus for multiple lines simultaneously).

Question 97 — Atari State Representation and CNN Role

(a) Describe a typical preprocessing pipeline for Atari frames.

Solution:

1. **Grayscale:** Convert RGB ($210 \times 160 \times 3$) to grayscale ($210 \times 160 \times 1$) to reduce dimensionality.
2. **Downsampling:** Resize to 84×84 pixels using bilinear interpolation.
3. **Normalization:** Scale pixel values from $[0, 255]$ to $[0, 1]$.
4. **Frame skipping:** Repeat each action for 4 frames, use max over last 2 to handle flickering.

(b) Explain why the state is often represented as a stack of the last k frames rather than a single frame.

Solution: A single frame is **Markov-incomplete**: it does not encode velocity or direction of motion. For example, a moving ball in Pong cannot be distinguished from a stationary ball from one frame alone.

Stacking the last $k = 4$ frames gives the agent information about motion (velocity, direction) without requiring a recurrent architecture. The state becomes approximately Markov: given the stack, the agent can infer short-term dynamics.

(c) Explain the role of a CNN for approximating $Q(s, a)$ from pixel inputs.

Solution: The stacked frames ($84 \times 84 \times 4$) are passed through a CNN that:

- Applies **convolutional filters** to detect local spatial features (edges, objects, textures) with **shared weights** – far fewer parameters than a fully-connected network on raw pixels.
- Exploits **translational equivariance**: a feature detected at one location generalizes to other locations.
- Stacks convolutional layers to detect increasingly complex spatial hierarchies.

The CNN outputs a feature vector fed to fully-connected layers, which produce one $Q(s, a)$ value per action.

(d) Discuss one alternative to CNNs for partially observable environments.

Solution: Recurrent architectures (LSTM/GRU): In partially observable environments (POMDP), the current observation may not fully determine the state (e.g., hidden obstacles, flickering). A recurrent network (e.g., LSTM) maintains a **hidden state** that aggregates information from the entire history of observations, approximating the belief state. This allows the agent to “remember” past observations and handle partial observability, unlike CNNs which only process the current (stacked) frame without memory.

Question 98 — Why Q-learning Should Converge: Proof Sketch

(a) State the Q-learning update and identify the “target” term.

Solution: Q-learning update for transition (s, a, r, s') :

$$Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + \alpha \underbrace{\left[r + \gamma \max_{a'} Q(s', a') \right]}_{\text{target } y}.$$

The **target** $y = r + \gamma \max_{a'} Q(s', a')$ is the Bellman optimality operator applied to Q at the observed next state.

(b) Explain the fixed-point equation that Q^* satisfies.

Solution: Q^* is the unique fixed point of the Bellman optimality operator T :

$$Q^*(s, a) = (TQ^*)(s, a) = \mathbb{E} \left[r + \gamma \max_{a'} Q^*(s', a') \mid s, a \right].$$

Q-learning iteratively applies T in a stochastic manner; as the number of updates to each (s, a) pair $\rightarrow \infty$, Q converges to the unique fixed point Q^* .

(c) Show that the Bellman optimality operator is a contraction in the sup norm with factor γ .

Solution: For any Q, Q' and any (s, a) :

$$|(TQ)(s, a) - (TQ')(s, a)| = \gamma \left| \mathbb{E} \left[\max_{a'} Q(s', a') - \max_{a'} Q'(s', a') \right] \right|.$$

Using $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$ and taking expectations:

$$\leq \gamma \mathbb{E} \left[\max_{a'} |Q(s', a') - Q'(s', a')| \right] \leq \gamma \|Q - Q'\|_\infty.$$

Taking sup over (s, a) : $\|TQ - TQ'\|_\infty \leq \gamma \|Q - Q'\|_\infty$. Contraction with factor $\gamma < 1$. $\square$

(d) Outline how stochastic approximation arguments imply convergence to the fixed point.

Solution: Q-learning can be written as a **stochastic approximation** algorithm:

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha_t [(TQ_t)(s_t, a_t) - Q_t(s_t, a_t) + \text{noise}_t]$$

where noise_t is zero-mean (in expectation). Under the **Robbins–Monro conditions** on the learning rates α_t :

$$\sum_{t=0}^{\infty} \alpha_t = \infty \quad \text{and} \quad \sum_{t=0}^{\infty} \alpha_t^2 < \infty,$$

and given that every state-action pair is visited infinitely often, stochastic approximation theory guarantees convergence to the unique fixed point Q^* (since T is a contraction). This is the Watkins–Dayan convergence theorem.

Question 99 — Policy Evaluation as a Linear System

For a finite MDP with fixed policy π and discount $\gamma < 1$:

(a) Derive the matrix equation $V = R^\pi + \gamma P^\pi V$.

Solution: The Bellman expectation equation for V^π :

$$V^\pi(s) = \sum_a \pi(a|s) \left[\sum_{s'} P(s'|s, a) (r(s, a, s') + \gamma V^\pi(s')) \right].$$

Define: $R_s^\pi = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) r(s, a, s')$ (expected immediate reward under π) and $P_{ss'}^\pi = \sum_a \pi(a|s) P(s'|s, a)$ (transition matrix under π). Then:

$$V^\pi(s) = R_s^\pi + \gamma \sum_{s'} P_{ss'}^\pi V^\pi(s'),$$

which in matrix form is $V = R^\pi + \gamma P^\pi V$.

(b) Solve for $V = (I - \gamma P^\pi)^{-1} R^\pi$ and state conditions for invertibility.

Solution: Rearranging: $(I - \gamma P^\pi)V = R^\pi \Rightarrow V = (I - \gamma P^\pi)^{-1} R^\pi$.

Invertibility: $(I - \gamma P^\pi)$ is invertible when $\gamma < 1$. Proof: All eigenvalues λ of P^π satisfy $|\lambda| \leq 1$ (since P^π is a stochastic matrix). Thus eigenvalues of $(I - \gamma P^\pi)$ are $1 - \gamma\lambda$ with $|1 - \gamma\lambda| \geq 1 - \gamma|\lambda| \geq 1 - \gamma > 0$. All eigenvalues are nonzero, so the matrix is invertible.

Moreover, $(I - \gamma P^\pi)^{-1} = \sum_{k=0}^{\infty} (\gamma P^\pi)^k$ (Neumann series, converges for $\gamma < 1$).

(c) Discuss why direct inversion is infeasible for large state spaces and name two alternative approaches.

Solution: Direct inversion: Computing $(I - \gamma P^\pi)^{-1}$ costs $O(|S|^3)$ time. For Atari-style tasks with $|S| \sim 10^{10}$, this is completely infeasible.

Alternatives:

1. **Iterative policy evaluation:** Repeatedly apply $V \leftarrow R^\pi + \gamma P^\pi V$ until convergence. Costs $O(|S|^2)$ per iteration, converges geometrically with rate γ .
2. **Temporal Difference (TD) learning:** Estimate V^π from sampled trajectories using: $V(s) \leftarrow V(s) + \alpha[r + \gamma V(s') - V(s)]$. Only requires $O(1)$ work per sample; applicable to large/continuous state spaces with function approximation.

Question 100 — Dynamic vs. Static Optimization

(a) Contrast a static optimization objective $\min f(x)$ with the RL objective $\max_\pi \mathbb{E}[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1}]$.

Solution:

Aspect	Static optimization	RL objective
Decision variable	$x \in \mathbb{R}^n$ (a vector)	$\pi : S \rightarrow \Delta(A)$ (a policy function)
Objective	$f(x)$, a fixed function	$\mathbb{E}_\pi[\sum_k \gamma^k R_{t+k+1}]$, expectation over trajectories
Nature	One-shot, deterministic	Sequential, stochastic, time-extended
Uncertainty	None (in classical setting)	Stochastic environment, unknown transitions
Feedback	Gradient ∇f	Rewards sampled from unknown $P(s' s, a), r$

Key difference: RL optimizes over *policies* (functions mapping states to actions) in a stochastic sequential setting, whereas static optimization optimizes over vectors in a fixed known landscape.

(b) Write an optimization problem whose decision variable is a policy π rather than a vector x .

Solution:

$$\max_{\pi: S \rightarrow \Delta(A)} \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_0 \sim \mu_0, A_t \sim \pi(\cdot | S_t), S_{t+1} \sim P(\cdot | S_t, A_t) \right]$$

subject to: $\pi(a|s) \geq 0$ for all s, a ; $\sum_a \pi(a|s) = 1$ for all s .

The decision variable π is a function in an infinite-dimensional space (for continuous S, A) or a matrix of size $|S| \times |A|$ (tabular case).

(c) Give two concrete examples of how RL can be used to optimize or tune numerical algorithms.

Solution:

1. **Adaptive step size in gradient descent:** State = (iteration number, gradient norm, loss value); Action = step size α ; Reward = $-(\text{loss after step})$. An RL agent learns a step size schedule that adapts to the loss landscape faster than fixed or hand-tuned schedules.
2. **Pivot selection in Gaussian elimination:** State = current matrix (or its statistics); Action = choice of pivot element; Reward = (negative of) numerical error or condition number growth. An RL agent learns a pivoting strategy that reduces fill-in or improves numerical stability beyond standard partial pivoting.

(d) Explain why self-play is effective in chess and what “state, action, reward” correspond to.

Solution: **State:** Current board position (piece placement, side to move, castling/en passant rights).

Action: A legal move from the current position.

Reward: +1 for win, −1 for loss, 0 for draw (at the end of the game; intermediate rewards are 0).

Why self-play works: By playing against itself, the agent always faces an opponent at approximately its own skill level. This creates a curriculum that naturally scales difficulty: as the agent improves, so does the opponent. Unlike training against a fixed opponent (which could lead to overfitting), self-play generates an endless, ever-harder stream of training games. This is the key idea behind AlphaZero.